

Regulatory Frictions, Prescribing Behavior, and Unintended Consequences: Evidence from Gabapentin Policies

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Abstract

Regulations, information constraints, and administrative burdens are key drivers of provider behavior; we use recent state-level policies targeting gabapentin as a natural experiment to study these mechanisms in prescribing decisions. Gabapentin, an increasingly misused medication with rapidly expanding off-label use, has been subject to two distinct regulatory interventions: Drug-of-Concern (DOC) designation, which expands PDMP reporting without constraining prescribing, and Schedule V classification, which imposes binding prescribing and documentation requirements. Using a novel dataset linking MarketScan claims (2016–2023) with CDC multiple-cause-of-death records, and a stacked difference-in-differences framework, we document four main findings. First, DOC designation has no detectable effect on gabapentin prescribing. Second, Schedule V classification leads to a sustained decline in prescribing, reducing treatment rates by 0.2–0.3 percentage points (from a 3–4% baseline) within eight quarters, driven by reduced continuation among established users. Third, we find little evidence of harmful substitution: opioid prescribing does not increase, co-prescribing with opioids declines, and pregabalin substitution is modest and delayed. Fourth, Schedule V classification is associated with a meaningful reduction in gabapentin-involved mortality, including opioid-co-involved overdose deaths, with no corresponding change following DOC designation. Mechanism tests exploiting variation in PDMP mandates and patient histories show that neither salience nor information effects generate measurable changes in prescribing; instead, the behavioral response is concentrated in settings where Schedule V classification raises administrative hassle costs. Collectively, the results demonstrate that only binding regulatory frictions meaningfully reduce gabapentin use and associated harms, while monitoring-only policies are largely ineffective.

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1 Introduction

The U.S. opioid epidemic has intensified scrutiny of pain management and drug regulation, raising questions about the unintended effects of supply-side policies. Restricting one class of medications can spur shifts to others; for instance, constraining OxyContin access contributed to a surge in heroin use (Alpert et al., 2018). Amid these regulatory shifts, gabapentin – an anticonvulsant – has seen dramatic growth in use and although gabapentin is not federally scheduled, its rising misuse and frequent co-prescription with opioids have drawn considerable attention and concern (Powell et al., 2023; Lavin et al., 2021; Torrance et al., 2020; Pauly et al., 2020; Hah et al., 2018). Gabapentin can potentiate opioid effects and, in some cases, combinations have been lethal (Powell et al., 2023), prompting calls for tighter regulations and scheduling that is comparable to opioids with high potential for misuse. Given its increasing off-label use for a range of conditions beyond its original indications, regulating gabapentin may alter treatment pathways and affect downstream health outcomes. This paper examines the impact of state-level efforts to monitor and control gabapentin prescribing, focusing on misuse, potential substitution to both other medications and non-pharmacological treatments, and mortality.

Gabapentin is now commonly prescribed off-label for various conditions, including migraines, some mental health disorders, insomnia, and alcohol withdrawal symptoms (Peckham et al., 2018; Modesto-Lowe et al., 2019). It is also increasingly used as an opioid-sparing agent in chronic pain management (Peckham et al., 2018). However, the evidence supporting these uses is generally weak, given that gabapentin is found to provide modest benefits (if any) for non-epileptic pain or mood disorders (Peckham et al., 2018). This extensive off-label use has made gabapentin one of the most prescribed medications in the U.S. – ranking seventh overall – largely for indications beyond its FDA labels (Powell et al., 2024). These prescribing patterns have occurred amid growing reports of abuse liability: individuals with substance-use histories report that gabapentin can elicit opioid-like euphoria, dissociative sensations, and enhanced sedation, especially at higher doses (Smith et al., 2016). Recent work also finds that co-administration of gabapentinoids (such as gabapentin)¹ with opioids nearly doubles the hazard of opioid overdose compared to opioids alone (Bykov et al., 2020). Given these concerning trends, the FDA and CDC have issued warnings that gabapentin can cause serious, life-threatening respiratory depression, especially when combined with opioids or other

¹Gabapentinoids include gabapentin, pregabalin, and mirogabalin.

central nervous system depressants (Mattson et al., 2022; U.S. Food and Drug Administration, 2019). Using data from the CDC’s National Vital Statistics System mortality data, Figure A1 shows a stark increase in the number of deaths where gabapentinoids are listed as a cause of death, and notably, 80% of these also involved an opioid. There is also evidence of physiological dependence: case reports document that tolerance and withdrawal symptoms can occur upon abrupt discontinuation of gabapentin (Smith et al., 2016). Taken together, emerging evidence indicates that gabapentin, once considered relatively benign, may carry meaningful risks even at approved doses—particularly when used in combination with opioids or other sedating medications (Bykov et al., 2020; U.S. Food and Drug Administration, 2019).

The concerning rise in gabapentin prescribing we have outlined has led to broadly two state-level policy responses. One, some states have classified or upscheduled the drug as a Schedule V controlled substance—despite its unscheduled status federally—thereby imposing stricter rules on prescribing, such as refill limits and subject to monitoring through the state prescription drug monitoring program (PDMP), which are centralized electronic reporting systems that record dispensing of controlled substances, and in some instances other prescription drugs with emerging misuse potential, with the aim of improving information flow to prescribers, detecting high-risk use patterns, and reducing diversion (Buchmueller and Carey, 2018). Grauer and Cramer (2022) find that these policies are associated with a reduction in the days supplied of gabapentin among Medicare Part D patients. When examining the upscheduling of a specific opioid, Tramadol, Gupta et al. (2023) finds that this change leads to a substitution to other opioids and thus no net change in opioid prescribing. Therefore, it is unclear whether gabapentin upscheduling can lead to harmful substitution or other prescribing practices, including the use of opioids. A possible concern with a Schedule V designation, is that in addition to various restrictive opioid policies, that such regulation of gabapentin may reduced appropriate access to patients with need for this type of pharmaceuticals. Two, rather than formally up-scheduling gabapentin, some states have designated it as a *drug of concern*, requiring its monitoring through state prescription drug monitoring programs (PDMPs). These distinct policy responses allow us to examine how health care prescribing and utilization differ under a monitoring-only regime versus a stricter Schedule V classification. This distinction is crucial for policymakers and clinicians: although both interventions aim to prevent misuse, they differ significantly in their regulatory requirements and clinical implications. Evaluating how these policy interventions shape prescribing behavior,

treatment alternatives, and patient outcomes can inform state efforts to promote safe prescribing while maintaining access to effective care.

In this paper, we use MarketScan® data, a longitudinal panel of employer-sponsored insurance claims covering 75 million individuals, and the staggered timing of state Schedule V classification and, less stringent, drug-of-concern (DOC) classification of gabapentin to examine prescribing and patient outcomes within the healthcare system. We first consider all individuals in the claims data and examine (1) gabapentin prescribing itself and how it changes after scheduling; (2) prescribing of opioids and other analgesics, including the rate of co-prescribing gabapentin with opioids; and (3) potential substitution to alternative pain treatments or medications (e.g., opioids or pregabalin). Pregabalin is particularly informative in this context. Pregabalin, a closely related gabapentinoid, has been a Schedule V controlled substance from the outset. Although chemically similar to gabapentin, pregabalin has always been federally scheduled as a Schedule V substance. When states upschedule gabapentin, the two medications face comparable regulatory cost of prescribing, creating incentives for substitution. Thus, pregabalin provides a natural comparison medication for assessing whether clinicians shift toward alternatives they perceive as less risky or more administratively convenient. We also consider spillovers into physical and mental healthcare; for instance, whether patients shift toward nonpharmacologic management, such as physical therapy.

We then build on the conceptual model and findings of (Alpert et al., 2024), who disentangle the roles of hassle costs and information provision resulting from a mandatory-access PDMP (MA-PDMP), incorporating the potential effects of salience. We leverage the fact that we can follow individuals over time within our healthcare claims data we test for evidence that hassle cost, information, and salience influence changes in gabapentin prescribing. Importantly, we can test for differences in providers’ prescribing responses to DOC and Schedule V policy changes, based on patients’ history of gabapentin prescribing, distinguishing between existing gabapentin users and gabapentin-naive patients. We can then tease out the channels driving prescribing changes by exploiting variation in DOC and Schedule V designations across mandatory-access and voluntary-access PDMP² states and across patient types. In states that don’t require mandatory PDMP checks, the DOC label mainly serves to make the drug more visible or top-of-mind, rather than introducing new information or administrative burdens. Thus, any decline in gabapentin

²Mandatory-access PDMP policies require prescribers to consult the PDMP in certain circumstances before prescribing opioids and other DEA-scheduled drugs, compared to voluntary PDMPs where the prescriber can choose to consult the PDMP or not (Buchmueller et al., 2020).

prescribing after the DOC designation—when no PDMP mandate is in place—is likely driven by heightened salience. In states with mandatory-access PDMPs, the DOC designation does more than increase salience—it also improves informational transparency by making patient-level prescribing histories, including patterns of co-prescribing with other controlled substances, more readily accessible to providers. Following [Alpert et al. \(2024\)](#), we examine the impact of Schedule V (C-V) classification to test for hassle costs. This designation introduces fixed administrative burdens—such as mandatory PDMP checks, refill limits, and more frequent patient visits—that are likely to reduce prescribing on the extensive margin by discouraging treatment initiation or continuation for both existing and new patients.

Our results indicate that Schedule V classification of gabapentin leads to an immediate and lasting decrease in prescribing of gabapentin. We find no effect of the drug-of-concern designation on prescribing. We also find no effect on spillovers to opioid prescribing. Together, this finding suggests a decrease in problematic co-prescribing of opioids and gabapentin. We also use mortality data from the CDC National Vital Statistics System and find that Schedule V classification leads to a reduction in mortality involving both gabapentinoids and opioids, as these co-occurring types of deaths account for about 12% of all opioid overdose deaths. We also find evidence of a substitution towards prescribing for pregabalin and away from opioids. When testing the mechanisms driving gabapentin’s effects by examining differences between existing gabapentin users and gabapentin-naïve patients, we find no effect of DOC classification on gabapentin prescribing in voluntary-access PDMP states across both patient groups, suggesting no effect of salience on prescribing. Additionally, we find that in mandatory-access PDMP states, there is no effect on existing nor naïve gabapentin users, limiting the effect of information on prescribing. Lastly, consistent with previous work, we find that Schedule V classification decreases gabapentin prescribing across all types of patients.

2 Background and Conceptual Model

2.1 Regulatory Context

Unlike medications like prescription opioids, gabapentin was long perceived as having a low abuse profile. However, rising overdose deaths involving gabapentin, alongside evidence that it may potentiate opioid effects and respiratory depression ([Bykov et al., 2020](#); [Mattson et al., 2022](#)), have raised serious concerns. In response, several states

have adopted policies to increase oversight of gabapentin prescribing. These include: (1) monitoring-only approaches, wherein gabapentin is designated as a “drug of concern” and tracked in state prescription drug monitoring programs (PDMPs); and (2) formal scheduling, where the drug is reclassified as a Schedule V controlled substance under state controlled substances acts, triggering the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced recordkeeping. These regulatory responses differ significantly in both stringency (Table 1 summarizes prescribing restrictions by regulatory classification), and therefore their potential implications for provider behavior.

Federal and state agencies may initially designate medications like gabapentin as “drugs of concern” as a lower-cost, lower-burden regulatory response while evidence on misuse is still accumulating. This approach allows the drug to be monitored through Prescription Drug Monitoring Programs (PDMPs) without invoking the legal and administrative processes required for formal scheduling. Under the Controlled Substances Act (CSA), federal scheduling requires a multi-step evaluation involving the Food and Drug Administration (FDA) and Drug Enforcement Administration (DEA), including an eight-factor analysis of abuse potential, public health risk, and medical utility (21 U.S.C. §811). This process can be time-consuming, politically sensitive, and dependent on robust epidemiologic and toxicologic data. Moreover, scheduling a widely used medication like gabapentin, which has recognized therapeutic value for chronic pain and neurological conditions, raises concerns about limiting access for legitimate patients, increasing prescriber burden, and potentially driving substitution toward more dangerous alternatives. As such, designating gabapentin as a drug of concern provides a regulatory middle ground that signals risk and allows for data collection without triggering the full constraints of a controlled substance classification ([Karavolis et al., 2021](#); [Drug Enforcement Administration, 2020](#)). At the state level, this designation also offers more flexibility and responsiveness, as state PDMP boards or health departments often have more discretion to flag emerging substances of concern without legislative or rule making delays.

The staggered designation of Drug of Concern (DOC) or Schedule-V (C-V) controlled substance status across states and over time generate plausibly exogenous variation in prescribing environments that we use to examine how regulatory frictions and information affect provider behavior.

2.2 Related Literature

Our analysis builds on several strands of research on behavioral responses to health policy. First, we relate to work showing that policy signals can alter provider behavior through changes in perceived risk or scrutiny, even without new information—for example, following FDA black-box warnings. DOC designation or scheduling may increase prescriber perception of the risks associated with gabapentin use, thereby negatively impacting the prescriber’s perception of the net gain from the medication for the patient, and increasing a sense of more dire consequences if now found negligent through prescriber monitoring. Previous research on FDA drug withdrawals, FDA black-box warnings and Consumer Production Safety Commission non-prescription product (children’s toys) recalls shows that competitors in the same therapeutic class could experience negative spillovers in some settings (Parkinson et al., 2014; Collins et al., 2013; Freedman et al., 2012; Cawley and Rizzo, 2008). Such spillovers frequently arise from heightened salience: warnings elevate the perceived risk and potential legal consequences of prescribing (Beyer et al., 2013), and can lead clinicians to reassess the net benefits of continuing treatment. Similar responses have been documented in other regulatory contexts, including vaccine safety warnings, antibiotic stewardship interventions, and medical device recalls (Aiesh et al., 2023). We refer to this as a *salience channel*, in which regulatory labeling increases the perceived cost of prescribing.

Second, a large body of literature emphasizes the *informational role* of PDMPs in shaping prescribing (Bao et al., 2016; Meara et al., 2016; Buchmueller and Carey, 2018; Ellyson et al., 2022). PDMPs are intended to support the prevention and early identification of opioid use disorder and misuse by equipping prescribers with more comprehensive information on patients’ histories with federally controlled substances. A key operational distinction across states’ PDMPs is whether prescribers are legally required to consult the system before issuing a prescription. This requirement appears to matter: utilization rates hover around 50 percent in states where PDMP use is voluntary (The Pew Charitable Trusts, 2016), despite nearly universal PDMP adoption by 2020. A number of studies report limited average effects of PDMPs on opioid use (Paulozzi et al., 2011; Li et al., 2014; Brady et al., 2014; Moyo et al., 2017), but more recent research finds that effects are considerably larger when systems include a mandatory query component (Buchmueller and Carey, 2021; Bao et al., 2016; Dowell et al., 2016). Concerns remain about low utilization in states without a query requirement. Some providers cite technical barriers or limited guidance on interpreting PDMP data as reasons for non-use

(Haffajee et al., 2015). Nationally, only about half of physicians report using PDMPs in practice (Rutkow et al., 2015), which may help explain why earlier evaluations found minimal effects on prescribing or overdose rates (Paulozzi et al., 2011; Li et al., 2014; Brady et al., 2014; Moyo et al., 2017; Yarbrough, 2017). In response, several states have adopted mandates that legally require prescribers to consult the PDMP before dispensing controlled substances. These mandates carry penalties for noncompliance and may also reduce providers’ legal exposure in the event of misuse or overdose (Haffajee et al., 2015). In our setting, DOC designation combined with mandatory PDMP access generates a clear test for whether the provision of new information—as distinct from heightened salience alone—drives reductions in prescribing.

Third, we connect to work on *hassle* costs in health policy (Finkelstein et al., 2019; Alpert et al., 2024; Sacks et al., 2021). Schedule V classification introduces binding legal constraints such as refill limits, written prescription requirements, and enhanced penalties for noncompliance. Such administrative burdens can reduce utilization even when demand remains unchanged. In canonical models, ordeal costs screen out low-benefit users by raising participation costs. In our context, however, these administrative frictions apply uniformly to prescribers rather than selectively to patients, offering no screening value. Hence, ordeal and hassle costs are empirically indistinguishable, and we treat them jointly as a *hassle-cost* channel reflecting fixed administrative burden. Recent work reinforces this interpretation: Currie and Musen (2025) show that Medicaid prior authorization requirements for preschool antipsychotic prescribing reduce utilization substantially without detectable spillovers to non-targeted patients or privately insured children, strongly suggesting that administrative frictions—rather than information spillovers—are the dominant mechanism.

Fourth, our study contributes to the literature on unintended consequences of supply side constraints on prescribing of pain medication, particularly implications for harmful substitution behavior. While DOC may shift marginal prescribing practices, mandatory PDMP policies and particularly Schedule V designation may induce more substantial shifts in treatment choice by increasing the effective relative price of gabapentin. Clinicians may respond by substituting toward unscheduled or less-regulated alternatives (e.g., physical therapy, over-the-counter formulations), or toward similarly regulated but administratively familiar options such as tramadol or codeine combinations. Evidence from opioid markets suggests that such substitution is common: when hydrocodone combination products were up-scheduled to Schedule II in 2014, prescribing of tramadol

rose sharply and overall opioid dispensing did not decline (Gupta et al., 2023). Related work similarly shows that selective restrictions on one opioid can shift use towards other substitutes (Alpert et al., 2018).

These trade-offs mirror findings from earlier upscheduling policies. For instance, Gupta et al. (2023) find that upscheduling tramadol did not reduce total opioid prescribing due to substitution toward other opioids. Similarly, Grauer and Cramer (2022) show that gabapentin Schedule V designation is correlated with a reduction of days supplied in Medicare Part D, but downstream impacts on neither substitution nor mortality were evaluated. The current evidence is therefore incomplete on whether stricter classification improves patient safety or merely shifts prescribing patterns in unintended ways.

2.3 Conceptual Overview

Regulatory restrictions on gabapentin prescribing may influence its prescription dispensation through three related channels: (1) *risk salience*, in which regulatory labeling may heighten providers’ awareness of misuse risk, leading them to prescribe more cautiously even without new information. (2) *information revelation*, in which when combined with mandatory PDMP access, these policies reveal patient-specific controlled-substance histories, allowing prescribers to incorporate new information about potential patterns of misuse into their decisions. And, (3) *hassle costs*, where regulatory requirements such as documentation, refill limits, and mandatory PDMP checks impose administrative burdens that may deter (frequent) prescribing independent of clinical or informational considerations.

Together, these mechanisms provide a conceptual framework for understanding how state-level policy interventions can alter provider behavior through changes in perceived risk, available information, and the cost of compliance.

The next section develops a formal model that embeds these channels and derives empirical predictions summarized in Table 3.

3 Theoretical Framework

3.1 Provider Decision Problem

We model a provider’s decision to prescribe gabapentin to patient i as a tradeoff between expected therapeutic benefit, perceived risk of misuse, and the hassle cost imposed by administrative burdens of prescribing. Let the utility from prescribing be:

$$U_i = \alpha B_i - \mathbb{E}[C(r_i) \mid \mathcal{I}(s)] - h(s, \kappa),$$

where B_i denotes the expected clinical benefit to the patient, $C(r_i)$ is the perceived cost of prescribing that depends on the patient’s observed or inferred risk profile r_i , $\mathcal{I}(s)$ represents the information set available to the provider under policy environment s , and $h(s, \kappa)$ captures the hassle cost induced by administrative or regulatory burden of prescribing that depends on the state policy environment s and PDMP design features κ , such as mandatory access PDMP. The provider prescribes if $U_i \geq 0$, which defines an implicit threshold rule:

$$\tau(s, \kappa) = \frac{\mathbb{E}[C(r_i) \mid \mathcal{I}(s)] + h(s, \kappa)}{\alpha}.$$

A prescription is issued when $B_i \geq \tau(s, \kappa)$.

3.2 Impact of Stricter Prescribing Restrictions

Tighter controls on gabapentin prescribing—through (1) the Drug of Concern (DOC) designation and (2) Schedule V (C–V) classification—can shift prescribers’ decision calculus. Each channel operates through distinct mechanisms that affect the expected risk term $\mathbb{E}[C(r_i) \mid \mathcal{I}(s)]$ or the hassle cost of prescribing $h(s, \kappa)$.

1. Risk Salience.

DOC or C–V designation increases the salience of potential misuse and regulatory scrutiny, elevating the perceived likelihood of adverse outcomes. In states without mandatory-access PDMPs, the DOC designation primarily enhances *salience* rather than providing additional information or imposing new hassle costs. In this setting, the information set $\mathcal{I}(s)$ expands only through heightened attention, raising $\mathbb{E}[C(r_i) \mid \mathcal{I}(s)]$ uniformly across patients. The heightened salience increases the

prescribing threshold for all patients, particularly those who are gabapentin-naïve, leading to larger reductions in initiation relative to continuation.

2. Information Revelation.

In states with mandatory-access PDMPs, the DOC designation also enhances informational transparency by revealing patient-level prescribing histories, including co-prescribing of controlled substances. Providers now observe individual risk signals r_i more clearly and can update the perceived risk cost $C(r_i)$ accordingly:

$$\mathbb{E}[C(r_i) \mid \mathcal{I}(s)] = C(r_i) + \Delta C(r_i), \quad \text{where } \Delta C(r_i) > 0 \text{ for high-risk patients.}$$

This mechanism predicts larger prescribing reductions among patients flagged as high-risk, such as existing gabapentin users, or those with suspected misuse patterns, recorded in the PDMP.

3. Hassle Cost.

Schedule V (C–V) classification imposes additional administrative burdens, represented by $h(s, \kappa)$, such as mandatory PDMP queries, refill restrictions, and increased visit frequency. These fixed per-prescription costs discourage prescribing on the extensive margin (fewer treated patients) among both preexisting and new users.

4 Data and Methods

4.1 Policy Information

The dates of policy changes around gabapentin’s classification in state PDMPs differs between sources, including Grauer and Cramer (2022), Karavolis et al. (2021), and information from PDMP Assist, leading us to investigate each policy change through state laws and announcements. From 2017-2023, 7 states classified gabapentin as a Schedule V substance and 17 states and the District of Columbia classified it as a *drug of concern*. Figure 1 shows the geographic distribution of each policy across the US. The exact dates and sources for each policy change is shown in Table A1. Throughout our sample period, Nebraska required reporting of all prescription drugs in their PDMP, and thus will not be included in our analyses.

4.2 Health Insurance Claims

Our main measures of health-care use comes from the Merative MarketScan® Research Databases, a longitudinal panel of employer-sponsored insurance claims covering 2016–2023. MarketScan tracks employees and their covered dependents month-by-month, linking every paid claim or capitated encounter to eligibility files and basic demographics. The data span inpatient, outpatient, retail-pharmacy, and carve-out services and are supplied by roughly 350 large, self-insured employers. After applying our inclusion rules, the analytic sample contains about 75 million unique enrollees, including 53 million employee policy-holders. As shown in Table 2, an average state sees 414,262 enrollees per quarter. Gabapentin prescriptions were identified by collecting NDC codes from MarketScan’s RED BOOK³ which had product names containing any of the following: “gabapentin”, “horizant”, “gralise”, “neurontin”, “gabarone”, or “fusepaq fanatrex”. As shown in Table 2, an average of 12 patients per 1,000 are prescribed gabapentin in each state per quarter. Pregabalin prescriptions were identified with the same method using product names containing “pregabalin” or “lyrica”. Opioid prescriptions were identified using their ATC classification code “N02A,” and mapping this to all corresponding NDC codes, which were then flagged as such in the data. Nonpharmacologic therapy for pain procedures were identified using CPT codes identified in Table 3 of Lee et al. (2021) in both the outpatient and inpatient claims files.

Using the same NDC codes to identify gabapentin and pregabalin, we extract the same estimates from the public use Medicaid State Drug Utilization Data (SDUD) files and Medicare Part D prescribing data. To calculate rates, we use state Medicaid and Medicare enrollee estimates from the Kaiser Family Foundation’s (KFF) State Health Facts tables “Health Insurance Coverage of the Total Population,” which provides estimates of individuals using each type of insurance from 2016-2024.

4.3 Mortality

Mortality data from 2010 to 2023 were obtained from the National Center for Health Statistics’ NVSS multiple cause-of-death files, which comprise the complete census of U.S. deaths. Overdose fatalities were identified via ICD-10 external cause-of-injury

³“The RED BOOK Table is a supplemental table that provides additional information about prescription drugs (for example, generic name, manufacturer, therapeutic class). Drugs are listed in this file by National Drug Code. The code is linkable to the Prescription Drug Claims (D) Table, so that selection of drug claims may be made by the categorical fields included in the RED BOOK,” MarketScan User Guide

codes X40–X44, X60–X64, X85, and Y10–Y14. Anticonvulsant involvement—including gabapentinoids and Z-drugs—was flagged using codes T42.6 (“other antiepileptic and sedative-hypnotic drugs”) and T42.7 (“unspecified antiepileptic and sedative-hypnotic drugs”) following [Powell et al. \(2023\)](#). Opioid-related deaths were defined by codes T40.0–T40.4 and T40.6 in accordance with [Ruhm \(2018\)](#). Decedents with both an anticonvulsant and an opioid code were classified as combination-involved overdoses. Although T42.6 and T42.7 do not explicitly name gabapentinoids or Z-drugs, these agents are encompassed within those categories in the standard ICD-10 classification. The Surveillance, Epidemiology, and End Results (SEER) population statistics are used to create mortality rates.

Figure [A1](#) plots the number of gabapentinoid-related deaths over time. Although, the total number deaths are relatively small, it is clear that these deaths have been increasing over time. Since 2010, the number of gabapentinoid related deaths have increased ten fold, with almost 90-percent of these deaths involving an opioid by 2023. Of the roughly 81,000 opioid-related deaths in 2023, approximately 12-percent of them involved a gabapentinoid.

We also see a similar trend in Figure [A2](#), which presents geographic trends in toxicology-confirmed gabapentin-involved poisonings reported to the fifty-five American Association of Poison Control Centers (AAPCC) and recorded in the National Poison Data System (NPDS) from 2011 to 2017.⁴ These maps visualize 3-digit ZIP code-level rates of Poison Help calls involving gabapentin exposures, adjusted for population and categorized using percentile cutoffs anchored to 2013 call volumes. Over this period, gabapentin-related poisoning rates increased in both incidence and geographic spread. This rise in non-fatal and fatal gabapentin-involved poisonings highlights an important public health concern, especially given that Poison Help call data reflect real-time toxic exposure reports and include cases beyond overdose deaths. As these confirmed poisoning episodes—over 60% of which originate from health care facilities—often serve as early indicators of misuse, they may be especially responsive to regulatory efforts like controlled substance scheduling or drug-of-concern designations. The observed trends reinforce the need to systematically evaluate how prescribing controls affect gabapentin use and misuse in both monotherapy and poly-substance contexts.

⁴Unfortunately, we do not have access to these data past 2017, and the first year of Schedule V designation is 2017.

4.4 Empirical Strategy

We exploit the staggered adoption of two distinct state-level policy interventions—(i) Schedule V classification of gabapentin and (ii) “drug of concern” designation requiring PDMP monitoring—to estimate their causal effects on prescribing and downstream outcomes. Our primary identification strategy is a stacked difference-in-differences (DiD) event-study framework, which pools cohorts of states adopting the policy in different years while flexibly controlling for calendar and cohort trends (Cengiz et al., 2019; Wing et al., 2024). Let s index states, t calendar quarter-year, and g the “cohort” equal to the quarter-year in which state s first implements the relevant policy. We estimate

$$Y_{g,s,t} = \alpha_s + \zeta_t + \eta_g + \sum_{\substack{k=-8 \\ k \neq -1}}^8 \beta_k D_{g,s,t}^{(k)} + \varepsilon_{g,s,t}, \quad (1)$$

where $Y_{g,s,t}$ is the outcome of interest (e.g., gabapentin prescriptions per 1,000 population, opioid co-prescribing rates, non-pharmacologic therapy visits, or overdose mortality), α_s , ζ_t , and η_g are state, quarter-year and cohort fixed effects, respectively. The indicator $D_{g,s,t}^{(k)}$ is equal to one when state s is in event-quarter-year k relative to its adoption quarter-year g (with $k = -1$ omitted), and zero otherwise. Standard errors are clustered at the state level to allow for arbitrary serial correlation (Bertrand et al., 2004). Coefficients β_k trace the dynamic response before and after adoption; small and insignificant estimates of β_k for $k < 0$ supports the parallel-trends assumption, while β_k for $k \geq 0$ capture the timing and persistence of treatment effects.

For ease of interpretation, we also estimate a collapsed stacked DiD model:

$$Y_{s,t} = \alpha_s + \zeta_t + \delta \text{Treat}_{s,t} + \mathbf{X}_{s,t} \psi + u_{s,t}, \quad (2)$$

where $\text{Treat}_{s,t}$ equals to one for all $t \geq g$ in treated state s , and zero otherwise. Here, δ measures the average post-policy change across treated states.

4.5 Empirical Tests of Mechanisms

After establishing the overall impact of tighter prescribing restrictions on gabapentin use, we next examine the mechanisms through which these policies operate. The theoretical framework in Section 3 suggests three primary channels: (i) increased perceived risk of gabapentin misuse (*salience*), (ii) additional patient-specific information revealed

through PDMP queries (*information*), and (iii) increased administrative burden associated with tighter prescribing controls (*hassle cost*). We conduct a series of robustness and heterogeneity tests to evaluate the three channels—salience, information revelation, and hassle cost—that may explain the observed relationship between tighter regulation of gabapentin and changes in its’ prescription dispensing. Each test exploits variation across policy environments, prescriber types, and patient subgroups.

4.5.1 Risk Salience (Risk Perception Channel).

A decline in prescribing following DOC adoption, even in states without access to new patient-specific information (via MA-PDMPs) or additional administrative requirements (via Schedule V reclassification), would indicate that providers respond to increased salience of perceived risk associated with gabapentin prescribing. We test this hypothesis using a difference-in-differences (DiD) model comparing prescribing and harm outcomes in states that labeled gabapentin as a DOC but did not operate a mandatory-access PDMP ($MA-PDMP = 0$) to those in never-treated states:

$$Y_{ist} = \alpha_i + \delta_s + \theta_t + \beta^{SAL} DOC_{st} + \varepsilon_{ist}, \quad (3)$$

where Y_{ist} denotes a prescribing outcome (e.g., the probability that patient i in state s and quarter t receives any gabapentin). A negative and statistically significant β^{SAL} would be consistent with a salience-driven reduction in prescribing, operating through heightened provider perception of misuse risk rather than through information revelation or administrative frictions.

Because initiation decisions are more discretionary than continuation of existing therapy, we expect heightened salience of gabapentin-related risks to have a larger impact on new starts rather than ongoing use. We test this by comparing effects for gabapentin naive vs existing gabapentin users by stratifying this model by each type. A larger β^{SAL} for the new users would be consistent with a stronger salience effect on initiation than continuation.

4.5.2 Information Effect (Patient-Specific Signal Channel).

When DOC designation coincides with a mandatory-access PDMP, providers obtain new patient-specific information on prior controlled-substance use. If this information reveals greater patient-specific misuse risk, prescribing should decline more sharply in

MA-PDMP states, especially for patients with a documented history of high-risk use. We test this by comparing changes in prescribing and harm outcomes between states that designate gabapentin as a DOC with and without MA-PDMP requirements, estimating:

$$Y_{ist} = \alpha_i + \delta_s + \theta_t + \beta_1^{INF} DOC_{st} + \beta_2^{INF} (DOC_{st} \times MA-PDMP_s) + \varepsilon_{ist}. \quad (4)$$

where $DOC_{st} \times MA-PDMP_s$ interacts the policy indicator with a binary variable equal to one in states with an MA-PDMP and equal to zero in those with only voluntary requirements. A negative β_2^{INF} would suggest that the DOC effect is amplified in MA-PDMP states. Again, we stratify by the existing and naive gabapentin users, where a larger decline for the existing patients would be consistent with providers incorporating newly revealed patient-level risk information into prescribing decisions.

4.5.3 Hassle-Cost (Administrative Burden Channel).

Finally, we test whether the additional administrative requirements associated with Schedule V (C–V) reclassification—such as refill limits and PDMP checks—increase the per-prescription hassle cost. We estimate:

$$Y_{ist} = \alpha_i + \delta_s + \theta_t + \beta^{HAS} CV_{st} + \varepsilon_{ist}, \quad (5)$$

where as above, Y_{ist} denotes a prescribing outcome. A negative and statistically significant β^{HAS} would indicate the hassle cost driving a reduction in prescribing through administrative frictions, on top of the salience and information still present.

We will again test the effects on continuation versus initiation by stratifying this model by gabapentin naive and existing gabapentin users. A larger β^{HAS} for the new users would be consistent with a stronger hassle cost effect on initiation.

Together, these tests provide evidence on the whether and how the three mechanisms - (i) risk-salience shifts in $\mathcal{I}(s)$, (ii) information-driven updates to perceived misuse risk $C(r_i)$, and (iii) administrative frictions reflected in $h(s, \kappa)$ - may contribute to the changes in gabapentin prescription dispensing following tighter prescribing regulations. Each regression complements the unified event-study and DiD models and provides mechanism-specific validation of the theoretical framework.

4.6 Descriptive Analysis

The relationship between gabapentin drug monitoring gabapentin prescribing is presented in Figure 2, which plots the number of patients with a gabapentin prescription (per 1,000) relative to 2016 over time for both drugs of concern states, Schedule V states, and those without a restriction. Initially, there is a slight decrease for Schedule V states, while drug of concern states and states without any gabapentin policy are on a similar trend. In 2019, there was a clear precipitous drop in prescribing in Schedule V states, which coincided with four of the seven states with such designation adopting this relatively stricter policy that year and all seven states having the designation by the end of this year. Conversely, drug of concern states seem to be on a similar trend throughout the sample period. Taken together, the trends show that gabapentin prescribing decreased relatively faster among Schedule V states. However, these results do not exploit variations in the timing of either policy. In the following sections, we run additional tests and examine the event-study estimates to provide more conclusive evidence of the effects of Schedule V and drug of concern policies.

5 Results

5.1 Effects of Gabapentin Regulation on Its Prescribing

Figures 3a and 3b present difference-in-differences event study estimates of the effects of state-level gabapentin regulation on prescribing. The outcome of interest is the share of patients with a gabapentin prescription, estimated in event time relative to the state-specific quarter of policy adoption (event time zero), with leads and lags centered around the quarter of implementation.

Figures 3a show the effects of drug-of-concern designation, which ensures that gabapentin prescribing is included in the state PDMP. We observe no statistically or substantively meaningful change in gabapentin prescribing following this classification. Pre-treatment estimates are flat and centered around zero, supporting the validity of the parallel trends assumption. The post-treatment estimates are close to zero, and the confidence intervals overlap throughout, indicating that providers showed little to no change in behavior. In this analysis, Schedule V states are included in the control group, however, results are unchanged when these states are excluded (see Figure B1).

We next examine the effect of Schedule V classification on gabapentin prescribing.

States that designated gabapentin as a drug of concern (DOC) are included in the control group, which is particularly appropriate given the null effects of DOC designation documented above. In sharp contrast to those null findings, Figure 3b shows that Schedule V classification, which imposes binding prescribing constraints, induces a sustained and statistically significant decline in gabapentin use beginning in the first quarter after implementation. The magnitude of the effect increases over time, reaching roughly 0.2–0.3 percentage points (from a baseline of 3–4%) by 6–7 quarters post-policy. These results indicate that controlled substance scheduling, unlike PDMP-only monitoring, meaningfully reduces prescribing. Figure B2 further demonstrates that these results are robust to excluding DOC states from the control group.

Given that our main results rely on private commercial insurance claims, we use aggregated state level data to estimate the same outcomes in Medicaid and Medicare prescribing. Figure 4a and Figure 4b show similar decreasing impacts of Schedule V classification on prescriptions per 1,000 enrollees.⁵ Consistent with our findings, the drug of concern classification did not show significant changes for these payers as well, shown in Appendix Figures B3a and B3b. The results in Medicare Part D prescribing aligns with prior literature studying the association of the policies with Medicare prescribing (Grauer and Cramer, 2022).

Other policies during this time period may have impacted prescribing of pain medications, notably must access opioid PDMP policies and medical or recreational marijuana. We control for opioid MA-PDMP policies, legalization of medical marijuana, legalization of recreational marijuana, and the beginning of sales for recreational marijuana. None of these policies altered the main negative and significant impact of decreasing gabapentin prescriptions following the Schedule V classification, as shown in Figure 5.

Together, these figures demonstrate that stricter regulatory designation via Schedule V classification has a more pronounced effect on prescribing behavior than monitoring alone. These findings highlight the importance of regulatory intensity in shaping provider responses to prescription drug policy.

⁵The public-use versions of these data are only available at the yearly level.

5.2 Substitution Responses to Prescribing Regulations

5.2.1 Substitution Toward Other Prescription Drugs with Misuse Potential

Reclassifying gabapentin as a Schedule V controlled substance brought new hurdles for prescribers, like limits on refills, extra paperwork, and requirements for electronic prescribing. These prescribing “costs” align gabapentin with other Schedule V substances, such as pregabalin, and with lower-schedule opioids (e.g., tramadol, codeine combinations), which may also be used for overlapping indications such as neuropathic or musculoskeletal pain. As a result, prescribers may be incentivized to shift away from gabapentin toward pharmacologic substitutes that now face similar administrative hurdles but may be perceived as more familiar, more potent, or more effective. To assess whether such substitution occurs, we examine trends in prescribing of pregabalin, opioids, and co-prescriptions involving gabapentin and opioids following Schedule V designation.

Figures 6 and 7 examine the potential substitution behaviors in response to the Schedule V designation of gabapentin, focusing on changes in opioids (broadly defined), co-prescriptions of gabapentin with opioids, and prescribing of pregabalin (a pharmacologic analog). Estimates are again generated using a stacked difference-in-differences event study approach under staggered treatment timing.

Figure 6a investigates substitution toward prescription opioids more broadly. Here, we find no significant increase in opioid prescribing following gabapentin scheduling. Point estimates remain close to zero and statistically indistinguishable from pre-trend values, providing no clear evidence of net substitution toward opioids. Figure 6b assesses changes in the rate of co-prescribing of gabapentin and opioids. Consistent with the overall decline in gabapentin prescribing observed earlier, we document a statistically significant and persistent reduction in co-prescriptions beginning in the first quarter post-policy. These results suggest that Schedule V classification effectively reduces high-risk combinations of gabapentin and opioids, but only through a reduction in gabapentin prescribing.

Figure 7 shows a modest but statistically detectable increase in the share of patients receiving pregabalin prescriptions beginning roughly three quarters after the policy is implemented, suggesting that some prescribers may substitute toward pregabalin, another Schedule V anticonvulsant, when gabapentin prescribing is restricted.

Taken together, these findings imply that Schedule V designation of gabapentin may reduce risky co-prescribing without inducing substitution toward more harmful alternatives like opioids. These results are robust to controlling for other pain related policies,

shown in Appendix Figure C5. However, the modest rise in pregabalin use highlights the possibility of therapeutic substitution toward other gabapentinoids, which may carry similar risks when co-used with opioids.

5.2.2 Substitution Toward Non-Pharmacological Pain Management

We next examine whether tighter prescribing controls on gabapentin induced substitution toward non-pharmacological pain management modalities, an important margin of adjustment if clinicians respond to rising prescribing frictions by shifting patients towards low-risk physical therapy, behavioral approaches, or other non-pharmacologic procedures.

Figure 8 presents event-study estimates for Schedule V classification. We observe a gradual and statistically significant increase in non-pharmacologic pain management beginning shortly after implementation, with effects growing over time. By 6–8 quarters, estimates are positive, persistent, and meaningfully above pre-policy trends. This pattern is consistent with prescribers reallocating some patients toward non-pharmacologic treatment when gabapentin prescribing becomes administratively more burdensome.

Figure C4 provides the corresponding estimates for Drug-of-Concern (DOC) designation. DOC status ensures that gabapentin prescribing is reported to the PDMP, but the mechanism through which this matters varies across states. In states with no PDMP-access mandate, DOC primarily raises salience without requiring provider review; in states with mandatory PDMP checks, DOC could increase both informational transparency and administrative workload. Consistent with the earlier finding that DOC classification has no detectable effect on gabapentin prescribing, Figure C4 shows no systematic change in non-pharmacologic management. Estimates remain close to zero and confidence intervals overlap substantially throughout the event window, indicating that, whether through salience alone or through salience plus information, DOC policies do not meaningfully shift treatment toward non-pharmacologic alternatives.

Taken together, these results reinforce a central pattern in our analyses: strong regulatory actions that impose binding constraints on prescribing (Schedule V classification) generate measurable changes in downstream treatment choices, whereas policies that alter reporting but may not necessarily increase prescribing costs (for instance, DOC designation, particularly in absence of MA-PDMP) do not. The substitution toward non-pharmacologic pain management under Schedule V classification is modest in magnitude, perhaps indicative of prescribers shifting some patients away from pharmacologic

treatment as prescribing costs rise.

5.3 Impact on Gabapentin Involved Adverse Events

In Figure 9, we present stacked difference-in-differences event study estimates of the impact of the Schedule V reclassification of gabapentin on mortality involving gabapentinoids. Panel (a) displays estimates for all deaths involving a gabapentinoid, while panel (b) narrows the outcome to those deaths involving both a gabapentinoid and an opioid.

Across both panels, we observe a notable post-controlled substance designation decline in mortality rates, with a relatively stable pre-trend and a clear downward shift in the quarters following the Schedule V classification (denoted by the vertical dashed line). This pattern suggests a substantial reduction in fatal adverse events temporally aligned with the regulatory intervention.

Specifically, the average treatment effect on the treated (ATT) is -0.109 (panel a) and -0.106 (panel b), both statistically significant at the 5 percent level. These values correspond to approximately a 10 percent reduction in gabapentin-involved mortality rates, measured per 100,000 deaths, relative to the pre-policy mean. The effect is particularly salient in panel (b), which reflects the high-risk subset of cases involving concurrent opioid use, consistent with clinical concerns about the potential of gabapentin to increase opioid toxicity.

The attenuation in mortality is gradual but persistent, suggesting that the Schedule V designation may have had a sustained deterrent effect on risky polypharmacy use or prescribing behavior. While confidence intervals remain relatively wide, especially at the end of the event-time window, the consistent negative trajectory suggests a meaningful decline attributable to tighter prescribing restrictions on gabapentin through its controlled substance designation.

In contrast, Appendix Figure A11 reports estimates from analogous event studies for nonfatal adverse events involving gabapentinoids following the classification of Schedule V. In contrast to significant reductions in mortality, the ATT estimate for nonfatal events is precisely estimated at zero ($ATT = < 0.000$), and no discernible post-policy trend. This null result suggests that while Schedule V classification may be effective in mitigating the most severe outcomes, such as fatal overdoses, it may have limited impact on the broader set of less severe adverse events.

Further comparison with Appendix Figures A8–A10, which present analogous estimates for states that designated gabapentin as a DOC, a weaker regulatory action,

reinforces the importance of regulatory intensity. Panel (a) of Figure A8 shows no meaningful reduction in deaths involving gabapentinoids, and Figure A9 similarly shows no decline in gabapentinoid + opioid-involved fatalities. In fact, estimated ATTs in both cases are small and statistically insignificant ($ATT = 0.037$ and 0.057 , respectively). Figure A10 further confirms a lack of meaningful effect of DOC designation on non-fatal adverse events, with a near-zero ATT estimate (0.030) and wide confidence intervals throughout the post-policy period. These Schedule V classification results are also robust to controlling for other pain related policies, as shown in Appendix Figure D5.

The contrast between Figures 6 and A8–A10 underscores the importance of the regulatory intensity of policy interventions. While DOC classifications may serve as informational tools, our findings suggest that they are insufficient to meaningfully alter fatal or non-fatal adverse event trajectories. In contrast, the Schedule V designation—accompanied by stricter prescribing and dispensing controls—appears to exert a measurable and sustained impact on high-risk use and mortality.

Together, these findings support the hypothesis that regulatory scheduling of gabapentin may be effective in mitigating its most severe public health risks, particularly when co-used with opioids. This is consistent with broader efforts to constrain the supply-side drivers of polypharmacy-related overdose deaths.

5.4 Heterogeneous Effects by Prior Gabapentin Use

To examine how effects differ for patients initiating versus continuing gabapentin therapy, we stratify the continuously-enrolled MarketScan sample (2,851,760 individuals enrolled in all quarters from 2016 through 2023) into three cohorts based on prescription activity in the baseline year, 2016:

- **Gabapentin users:** individuals with at least one gabapentin prescription in 2016 (68,525)
- **Gabapentin naive:** individuals with no gabapentin prescription in 2016 (2,783,235)
- **Broad naive:** individuals with no prescriptions for gabapentin, pregabalin, nor opioids in 2016 (2,008,185)

The gabapentin naive cohort includes all individuals in the broad naive group, but also those who may have used pregabalin or opioids in 2016. The broad naive group represents individuals with no prior exposure to either gabapentinoids or opioids and

therefore provides a clean sample of new potential entrants into medications with these restrictions.

We estimate the specification in Equation 2 separately by group to examine prescribing of gabapentin, pregabalin, opioids, and other mental-health medications, as well as emergency department (ED) visits. Mortality cannot be observed in MarketScan, so ED visits serve as our measure of severe adverse outcomes.

Figure 10 presents the results. All groups experienced declines in gabapentin prescribing, with the largest and only significant effects among baseline gabapentin users, indicating substantial reductions in continuation. While statistically insignificant, there are negative average treatment on the treated estimates among naive groups, demonstrating potential evidence of reduced initiation into gabapentin use. Consistent with a hassle-cost mechanism, pregabalin prescribing increases significantly for both the existing gabapentin user and gabapentin-naive cohorts, indicating therapeutic substitution when both medications become equivalently regulated. We also find increases in other mental-health drug prescribing in all cohorts, consistent with substitution for off-label anti-anxiety and antidepressant uses of gabapentin. There is again no evidence of potentially harmful substitution to opioids for any cohort.

ED visits insignificantly decline across all cohorts, suggesting that increased gabapentin controls do not worsen acute clinical outcomes and may reduce severe events among baseline users. Taken together, these results indicate meaningful reductions in both continuation and initiation of gabapentin therapy, substitution toward other analgesic and psychiatric medications, and no evidence of increased acute harms.

5.5 Empirical Tests of Mechanisms

To separately identify the three behavioral channels in the conceptual framework—risk salience, information revelation, and hassle costs—we estimate Equations (3)-(5) using the quasi-experimental comparisons outlined in Table 3. Each mechanism maps to a distinct empirical comparison. Salience is identified from states that adopt Drug-of-Concern (DOC) designation without a mandatory-access PDMP (MA-PDMP), information from the $\text{DOC} \times \text{MA-PDMP}$ interaction (a triple-difference comparing DOC states with and without mandatory PDMP access), and hassle costs from Schedule V classification relative to never-treated states. We estimate Equations (3)-(5) using the continuously enrolled cohort and report results separately for the full sample, existing gabapentin users, and gabapentin-naive individuals. The analysis is conducted on the continuously

enrolled cohort, with results presented for the full sample, existing gabapentin users, and gabapentin-naïve individuals. Table 4 summarizes these estimates.

Salience. Panel A isolates cases where gabapentin is classified as a Drug of Concern (DOC) in states without a mandatory-access PDMP ($\text{MA-PDMP} = 0$), a setting in which the policy elevates salience but does not alter information or administrative burden. Across all specifications, estimated effects are small and statistically indistinguishable from zero. For the full sample, the coefficient is -16.31 ($\text{SE} = 10.88$); for existing users, 9.913 ($\text{SE} = 8.081$); and for gabapentin-naïve patients, 0.628 ($\text{SE} = 0.649$). These patterns indicate that salience alone does not materially alter prescribing behavior, consistent with evidence that PDMPs lacking mandatory-use provisions have limited influence on provider behavior.

Information. Panel B evaluates whether increased information availability through MA-PDMP infrastructure—captured by the interaction of DOC with the mandatory-access requirement—changes prescribing. Again, point estimates remain small and statistically insignificant across all groups. For the full sample, the coefficient on the information term is -1.032 ($\text{SE} = 0.623$); for existing users, -5.351 ($\text{SE} = 8.083$); and for naïve users, -1.234 ($\text{SE} = 1.141$). Although the negative coefficient among existing users is directionally consistent with the model’s prediction of information-driven adjustments, none of the estimates are statistically different from zero. These findings suggest that incremental information gained through PDMP access alone is insufficient to meaningfully shift prescribing behavior.

Hassle Costs. Panel C examines the effect of Schedule V classification, the policy that directly increases administrative burden. Here the effects are economically meaningful and statistically significant. For the full sample, the Schedule V estimate is -2.194 ($\text{SE} = 0.890$), a reduction of roughly 18 percent relative to the pre-policy mean. Among existing users, the decline is substantially larger: -17.20 ($\text{SE} = 5.732$), significant at the 1 percent level, indicating that continuation decisions are particularly sensitive to administrative frictions. For gabapentin-naïve individuals, the coefficient is negative but small and statistically insignificant (-0.138 , $\text{SE} = 0.734$), consistent with more modest impacts on initiation behavior.

Overall, the pattern of results provides clear empirical support for the importance of hassle costs as the operative mechanism. Policies that elevate salience or enhance

information flows do not generate measurable changes in prescribing, whereas regulatory interventions that impose binding administrative frictions produce large and precisely estimated declines concentrated among continuing users.

6 Discussion and Conclusion

This study provides novel evidence on the prescribing consequences of increased regulation of gabapentin, a drug that has seen substantial off-label growth in recent years and has become increasingly associated with misuse, particularly in combination with opioids. Leveraging the staggered implementation of state-level policies classifying gabapentin as either a drug of concern (DOC) or as a Schedule V controlled substance, we examine how prescribers respond to regulatory friction of varying intensity.

Our findings reveal that classifying gabapentin as a DOC, which entails inclusion in PDMPs but does not restrict prescribing, had little to no impact on prescribing rates. In contrast, Schedule V classification, which introduces stricter controls and documentation requirements, resulted in a sustained and statistically significant reduction in the share of patients receiving gabapentin prescriptions. These patterns suggest that regulatory designations must impose real prescribing costs to generate meaningful behavioral change, a result broadly consistent with prior work examining the rescheduling of opioids such as hydrocodone and tramadol ([Gupta et al., 2023](#)).

In assessing potential unintended consequences, we find limited evidence of clinically concerning substitution behavior. There is no detectable increase in prescribing of opioids, and co-prescribing of gabapentin and opioids declines following scheduling. However, we do observe a modest increase in pregabalin prescribing, a pharmacologic analog of gabapentin that was already federally classified as Schedule V. This is consistent with economic models of substitution, in which prescribers shift to alternative treatments facing similar regulatory barriers when relative prescribing costs are equalized. While such substitution may not carry the same overdose risk as opioid substitution, it still merits clinical scrutiny given emerging evidence on the misuse potential of gabapentinoids.

Our findings contribute to a growing literature evaluating supply-side drug policy interventions in settings where therapeutic and abuse-prone substitutes exist. They underscore the importance of considering the behavioral response of prescribers when designing regulatory interventions, especially for drugs that occupy ambiguous clinical and regulatory space. From a policy perspective, our results suggest that regulatory

classifications short of scheduling may be insufficient to curb use in high-risk contexts, and that policies must weigh potential substitution channels even when targeting non-opioid therapies.

Future work should assess patient-level outcomes, such as pain management effectiveness, adverse events, and healthcare utilization, to more fully characterize the clinical trade-offs of gabapentin scheduling. More broadly, as regulators continue to evaluate gabapentinoids and other off-label therapies, our study provides a framework for anticipating how prescribers may respond to differential policy levers and the resulting implications for public health.

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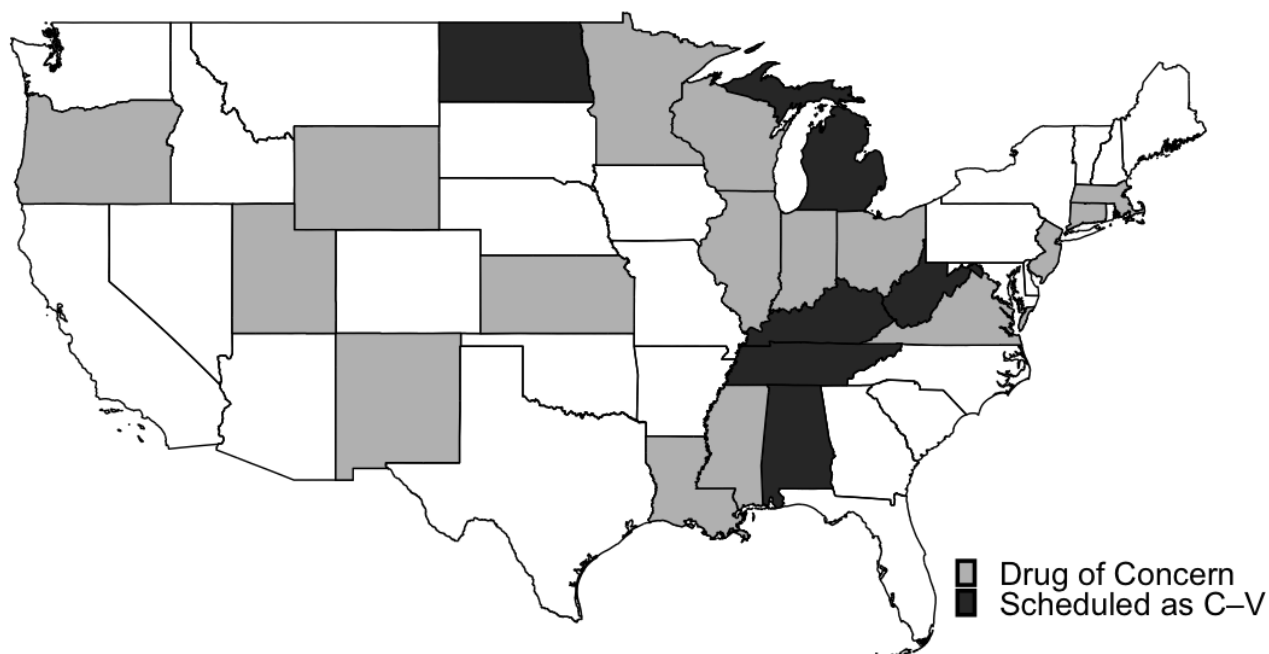
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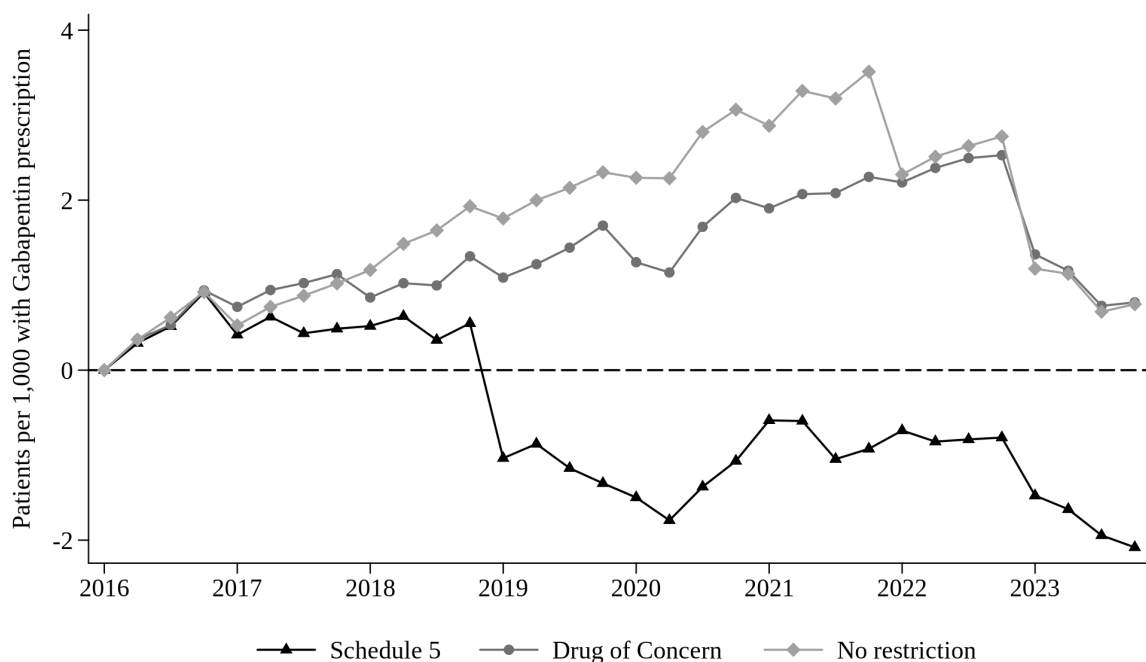
Figures

Figure 1: Policy Variation Across States



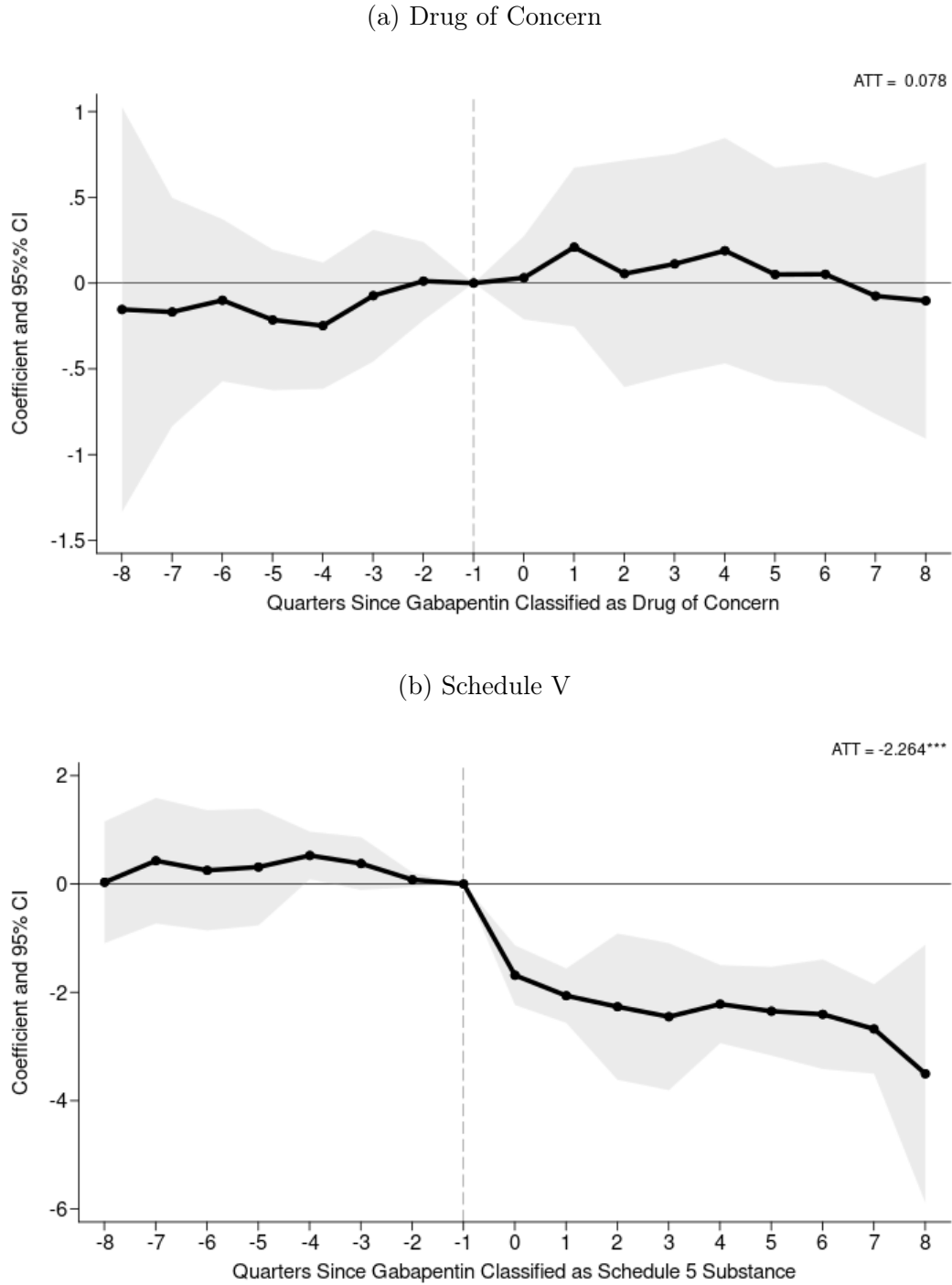
Notes: Map of state policies restricting prescriptions of Gabapentin by policy type as of December 2023. Drug of Concern policies add the drug to the state PDMP monitoring system without invoking the legal and administrative processes required for formal scheduling. Schedule-V policies trigger the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping.

Figure 2: Gabapentin Prescriptions by State Restriction Policy



Notes: Quarterly gabapentin prescriptions per 1,000 patients in MarketScan® commercial insurance claims data by state policy of ever classifying Gabapentin as a Schedule V controlled substance, ever as a Drug of Concern, or no restriction from 2016-2023. Drug of concern policies add the drug to the state PDMP monitoring system without invoking the legal and administrative processes required for formal scheduling. Schedule-V policies trigger the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Rates are normalized to 2016, the first year of our sample period. Rates without normalization are presented in Appendix Figure A3.

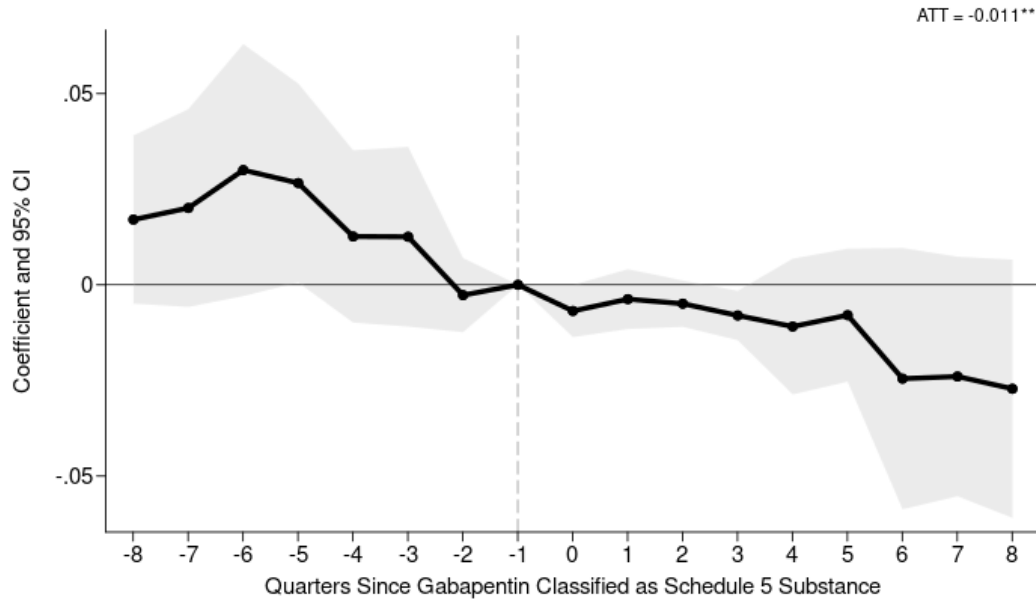
Figure 3: Gabapentin: Drugs of Concern vs. Schedule V Classification



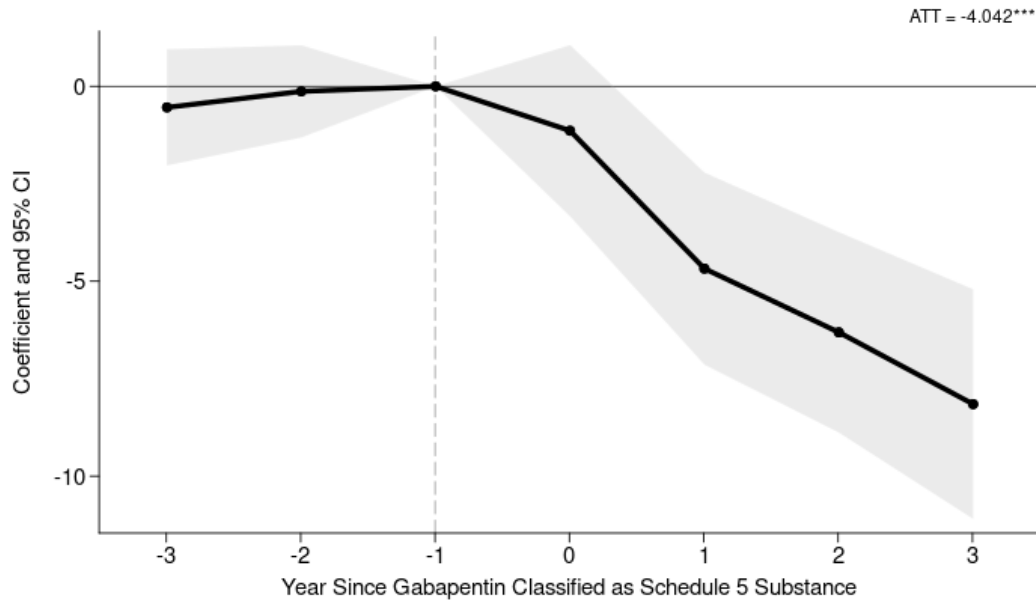
Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of policies restricting Gabapentin prescribing on Gabapentin prescription rates for patients in MarketScan® commercial insurance claims data 2016-2023. Estimates for the 8 quarters before and after the policy change. (a) Change in rate of patients per 1,000 with a Gabapentin prescription following classification in state to Drug of Concern, and thus included in the PDMP. Control group is never treated and Schedule V states. (b) Change in rate of patients per 1,000 with a Gabapentin prescription following classification in state to Schedule V controlled substance, triggering the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Control group is never treated and DOC states.

Figure 4: Gabapentin: Schedule V Classification - Public Payer

(a) Medicaid

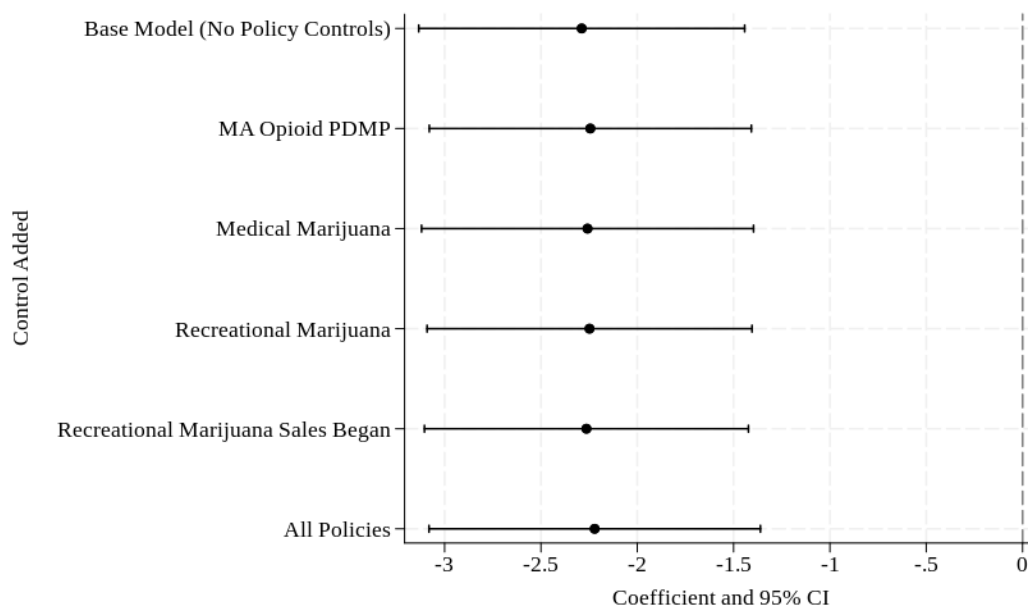


(b) Medicare



Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate impacts of classifying Gabapentin as a Schedule-V controlled substance on Gabapentin prescription rates for patients using publicly funded insurances: Medicaid and Medicare. Being a schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Control group is never treated and DOC states. (a) Change in rate of rate of Medicaid enrollees per 1,000 with a Gabapentin prescription following Schedule-V classification in state. Estimates for the 8 quarters before and after the policy change. (b) Change in rate of rate of Medicare Part D enrollees per 1,000 for an average quarter each year with a Gabapentin prescription following Schedule-V classification in state. Estimates for the 3 years before and after the policy change.

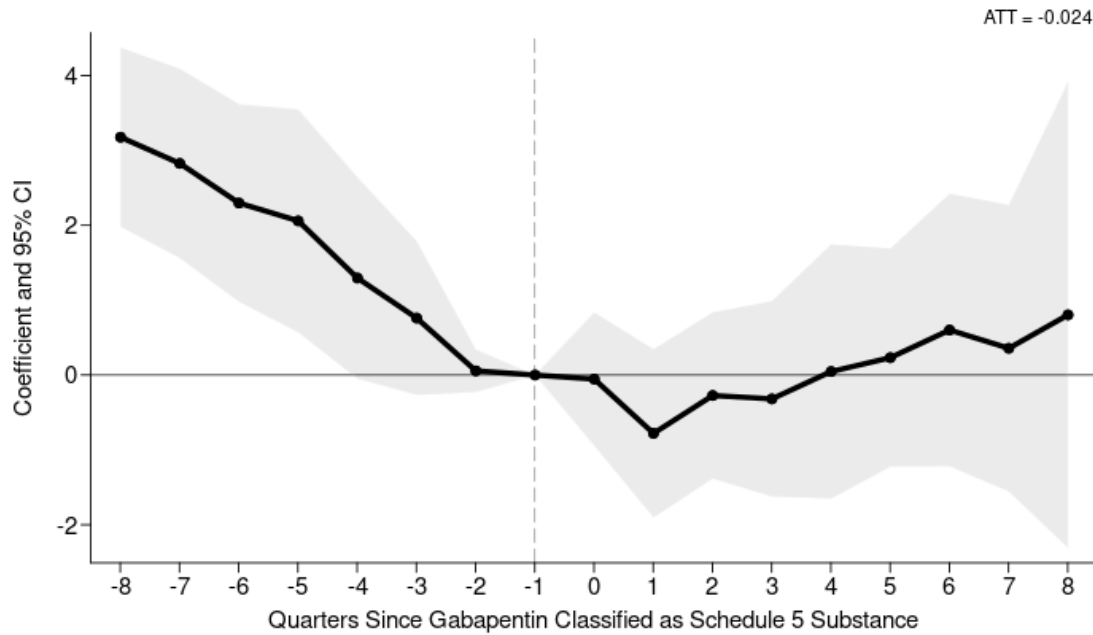
Figure 5: Gabapentin: Schedule V Classification, Other Policies Robustness Check



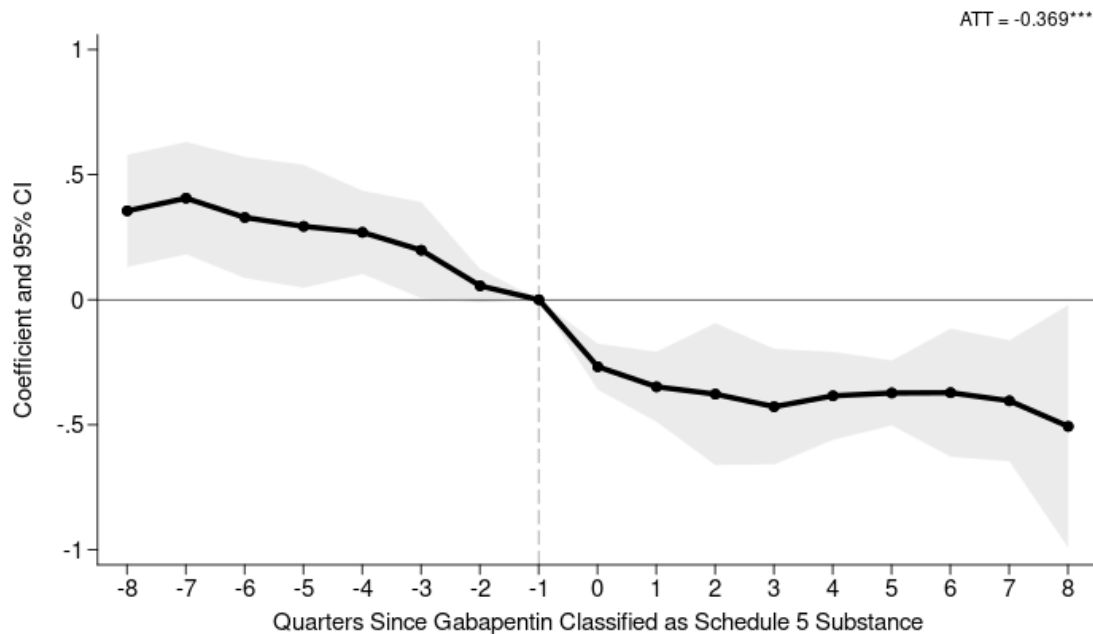
Notes: Average treatment on the treated effect (ATT) results using a stacked difference-in-differences model from Equation 2 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on Gabapentin prescription rates per 1,000 patients in MarketScan® commercial insurance claims data 2016-2023. Estimates based on the 8 quarters before and after the policy change. Control group is never treated and DOC states. Being a schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Each estimate below the base model includes control variables for other policies which may impact prescribing of gabapentin via its substitutes. MA Opioid PDMP policies require prescribers of opioids to consult the state PDMP prior to issues prescriptions. Medical Marijuana policies legalize the medical use and sale of marijuana and products which contain it. Recreational Marijuana policies legalize the use and sale of marijuana and related products without the need for medical reasoning nor a prescription. Recreational sales in certified establishments often began after formal recreational legalization, making these later dates the timing when individuals may have purchased the products from a shop. The bottom estimate includes controls for all policies.

Figure 6: Schedule V Classification Effects on Opioid Prescribing

(a) Opioids

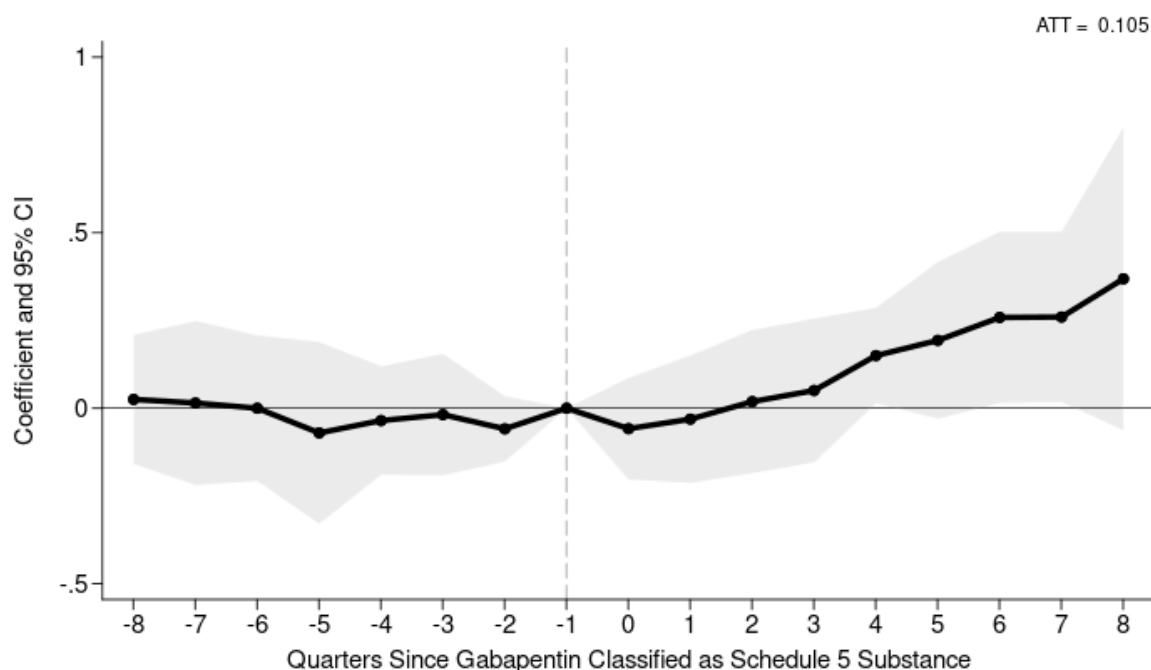


(b) Co-prescribing Opioids and Gabapentin



Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on prescription rates for patients in MarketScan® commercial insurance claims data 2016-2023. Being a Schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Estimates for the 8 quarters before and after the policy change. Control group is never treated and DOC states. (a) Change in rate of patients per 1,000 with an opioid prescription following Schedule V classification of Gabapentin in state (b) Change in rate of patients per 1,000 co-prescribed Gabapentin and an opioid following Schedule V classification of Gabapentin in state.

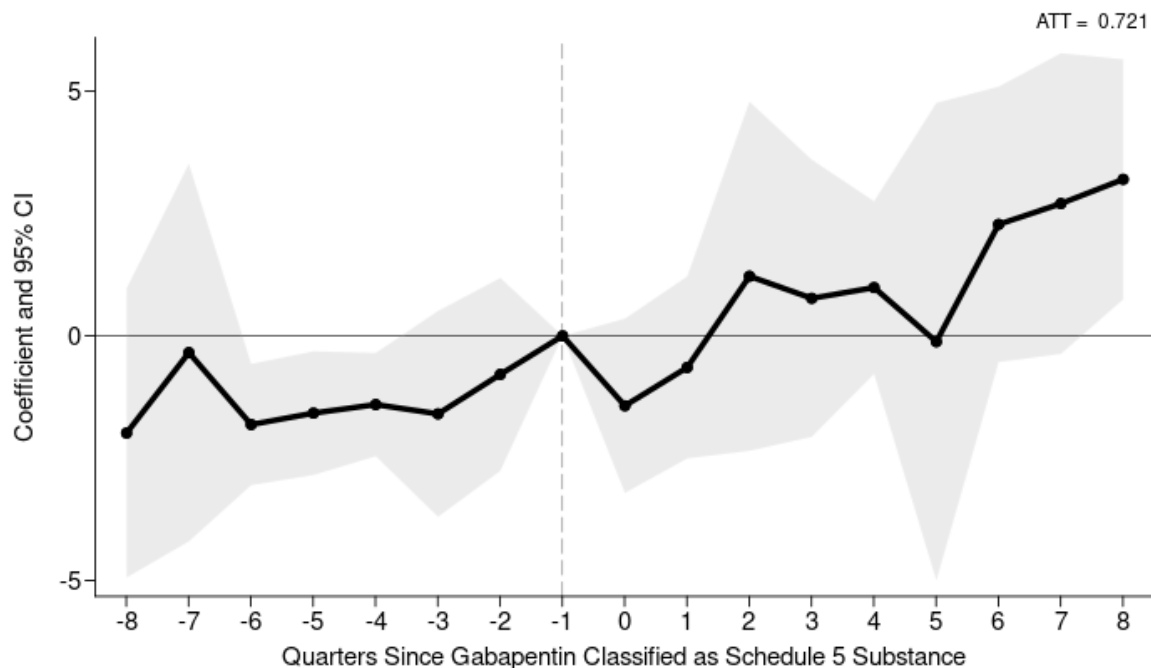
Figure 7: Schedule V Classification - Pregabalin Impact



Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on prescription rates of Pregabalin – a potential substitute for Gabapentin – per 1,000 patients in MarketScan® commercial insurance claims data 2016-2023. Being a Schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Estimates for the 8 quarters before and after the policy change. Control group is never treated and DOC states.

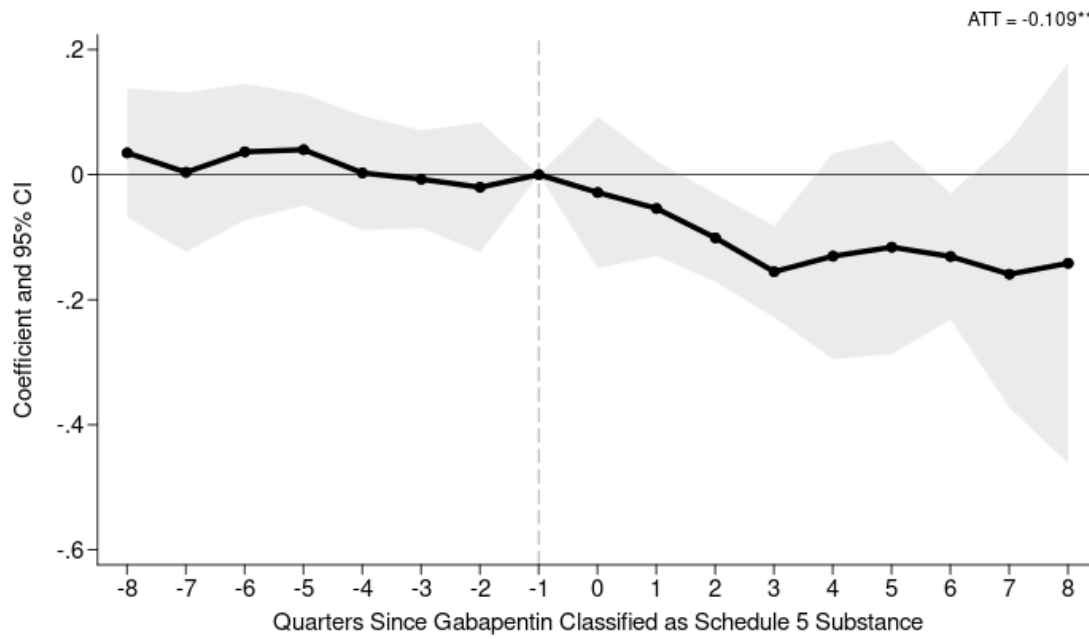
Policy Impacts on Non-Pharmacological Pain Management

Figure 8: Schedule V Classification - Non-Pharmacological Pain Management

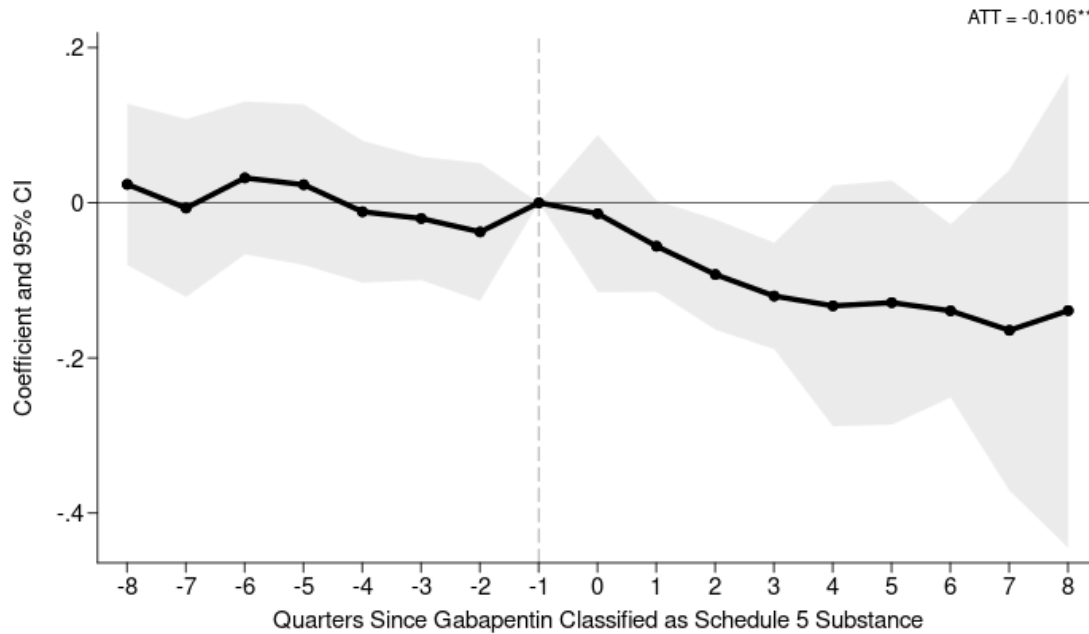


Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on rates of patients per 1,000 with a non-pharmacological pain management procedure among patients in MarketScan® commercial insurance claims data 2016-2023. Being a Schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Estimates for the 8 quarters before and after the policy change. Control group is never treated and DOC states.

Figure 9: Effects of Schedule V Classification on Mortality Involving Gabapentin



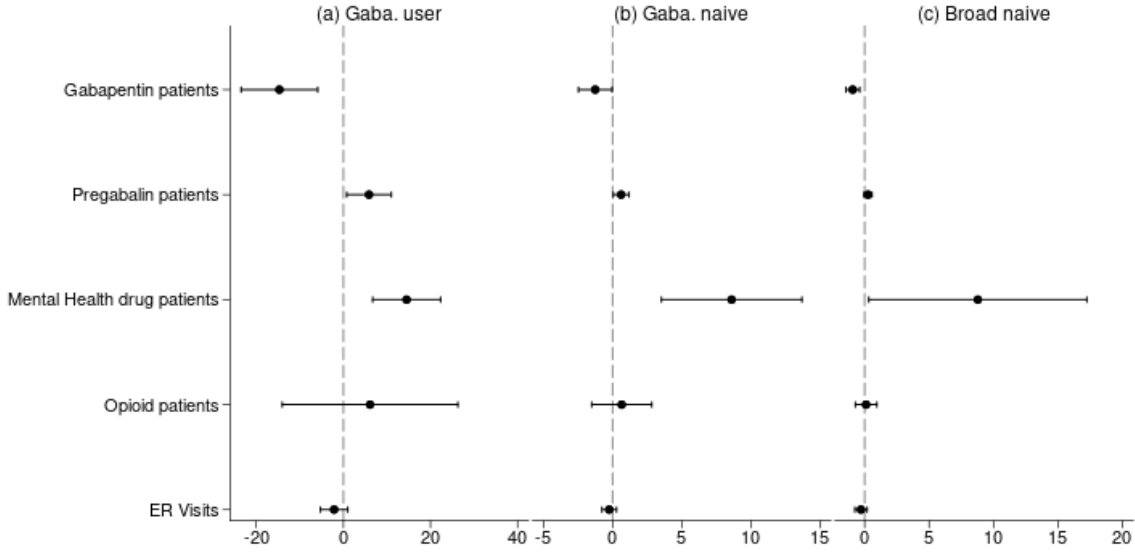
(a) Gabapentinoids



(b) Gabapentinoids + Opioids

Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on mortality rates from the National Vital Statistics System (NVSS) from 2016-2023. Being a Schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Estimates for the 8 quarters before and after the policy change. Control group is never treated and DOC states. (a) Change in mortality rate per 100,000 of state population including anticonvulsants following Schedule V classification of Gabapentin in state. (b) Change in mortality rate per 100,000 of deaths involving anticonvulsants and opioids following Schedule V classification of Gabapentin in state.

Figure 10: Schedule V Classification - Impacts by Existing Gabapentin Exposure



Notes: The sample from MarketScan® commercial insurance claims data is subset to those continuously enrolled for each quarter from 2016-2023. Three groups are created based on their 2016 prescription records. Panel (a) shows results for patients who were existing gabapentin users, having a prescription in 2016. Panel (b) shows results for gabapentin-naive patients, who did not have a gabapentin prescription in 2016. Panel (c) shows results for a broader naive category of patients who did not have gabapentin, pregabalin, nor opioid prescriptions in 2016. Point estimates and confidence intervals for average treatment on the treated effect (ATT) results using a stacked difference-in-differences model from Equation 2 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on prescription and ER visit rates of patients per 1,000 patients in MarketScan® commercial insurance claims data 2016-2023. Results are shown for Gabapentin prescribing, potential substitutes: Pregabalin, opioids, and mental health drugs for anxiety and depression, and adverse outcome of ER visits. Being a Schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Estimates for the 8 quarters before and after the policy change. Control group is never treated and DOC states.

Tables

Table 1: Characteristics and Regulations: Tiers of Prescribing Restrictions.

Prescription drugs	Drug of Concern	Schedule V Controlled Substance (Criterion: medical use, dependence and misuse potential)
1. Little/ no known risk of abuse. E.g. Antibiotics	1. Minimal abuse potential E.g. mild narcotic cough medicines	1. Low abuse potential E.g. Benzos, tramadol
2. Requires valid prescription	2. Prescription mostly valid for 12 mo. [†]	2. Prescription mostly valid for 12 mo. [‡]
3. Electronic/ phone scripts allowed w/o patient-provider in-person contact		
4. Refill as authorized by practitioner		
5. Delegate prescribing allowed		
6. Not require prescriber DEA number		6. Other general restrictions: • Require prescriber DEA number • Prescriptions must include patient name, patient DOB, drug name, dosage, strength, days of supply and refills (if any) • Prescription bottle must carry FDA warning label prohibiting transfer • Volume restrictions on wholesale distribution • Locked storage in pharmacy • Criminal penalties for unlawful or unlicensed distribution, possession, and use (without valid prescription)
Increasing restrictions on prescribing		

[†]With state ability to specify alternative validity limits. [‡]For instance, Kentucky mandates that prescription be filled within 60 days.

Table 2: State-Quarter Summary Statistics

	N	Mean	St. Dev.
Patients enrolled	1552	414,262.4	476579.5
Gabapentin prescriptions per 1,000 patients	1552	12.47189	3.024941
Pregabalyn prescriptions per 1,000 patients	1552	2.196201	.8060852
Opioid prescriptions per 1,000 patients	1552	24.28274	7.187883
Co-prescriptions of gabapentin and opioids per 1,000 patients	1552	3.025954	.953004
Anticonvulsant involved mortality per 100,000	2824	.3439374	.5275725
Anticonvulsant and opioid involved mortality per 100,000	2824	.2765101	.4899691
Gabapentin non-fatal poisoning diagnoses per 10,000 patients	1552	.388167	.2268058
Non-pharmacological pain management per 1,000 patients	1552	56.87691	18.99491

Table 3: Empirical Mechanisms and Testable Predictions from the Model of Prescribing Under Policy-Induced Risk Salience and Hassle Costs

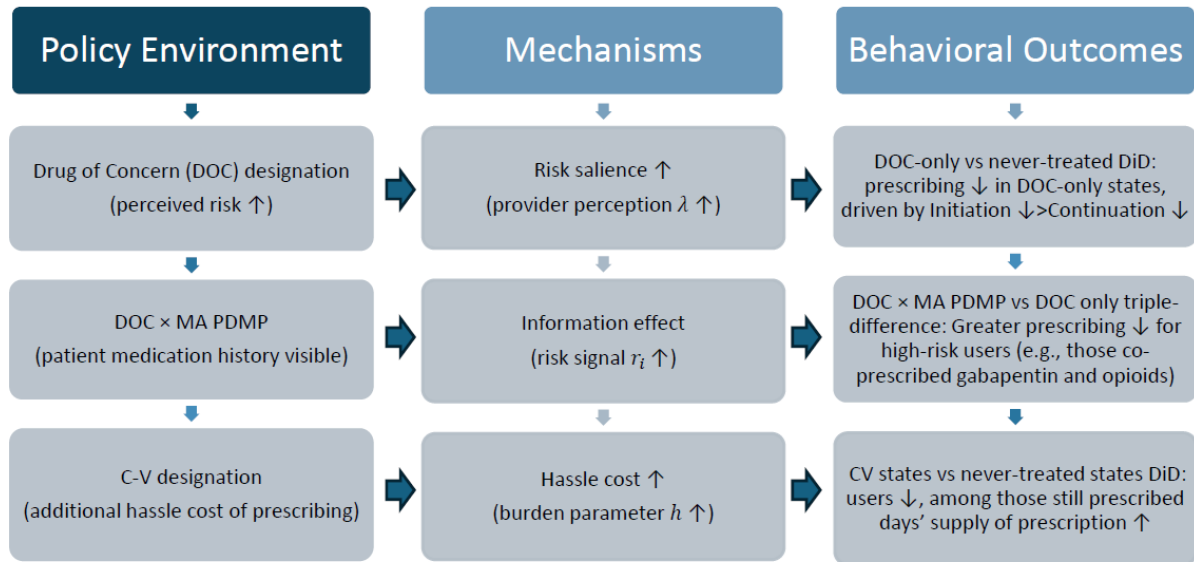


Table 4: Mechanism Check

Outcome: Gaba. pats per 1,000	(1) Full Sample	(2) Gaba-Existing	(3) Gaba-Naïve
A. Salience (No MA-PDMP + No S-V)			
Drug of Concern DiD	-16.31 (10.88)	9.913 (8.081)	0.628 (0.649)
	24,608 0.906	12,128 0.980	12,128 0.980
B. Information Effect (No S-V)			
Drug of Concern DiD	0.320 (0.499)	10.01 (7.961)	0.641 (0.644)
DoC DiD * MA-PDMP	-1.032 (0.623)	-5.351 (8.083)	-1.234 (1.141)
Observations	24,960	12,480	12,480
R-squared	0.906	0.980	0.979
C. Hassle Cost (No DoC)			
Schedule-V DiD	-2.194** (0.890)	-17.20*** (5.732)	-0.138 (0.734)
Observations	1,920	960	960
R-squared	0.008	0.982	0.978

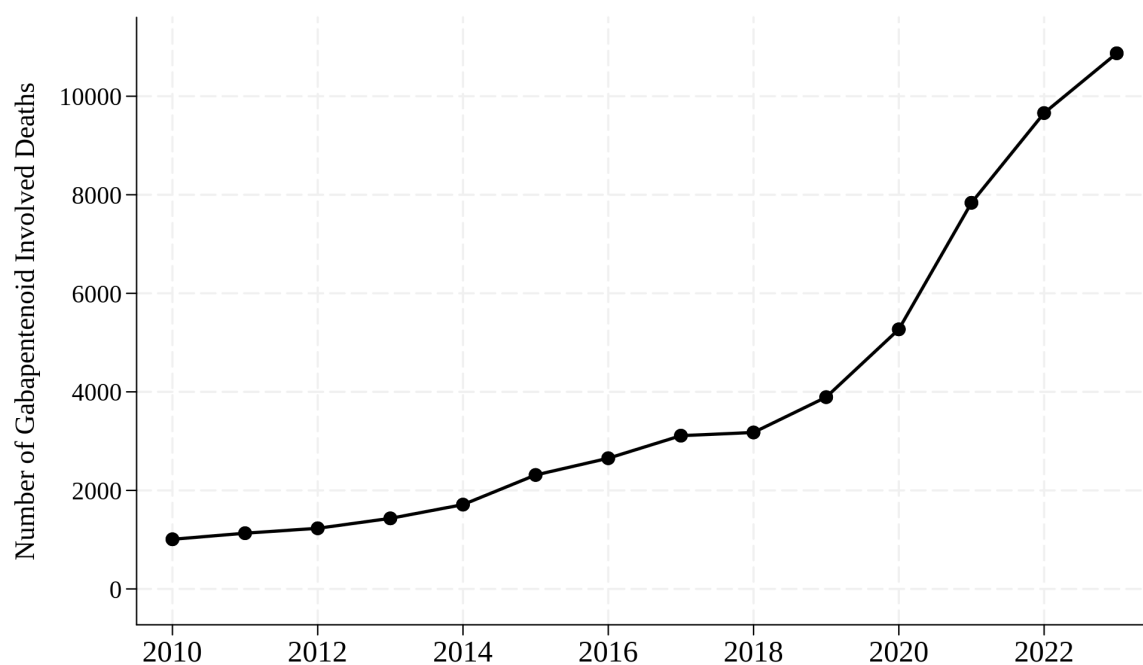
Notes: Robust standard errors in parentheses. For each panel, column (1) represents the full cohort sample of continuously enrolled patients from 2016-2023, column (2) is the subset who had a gabapentin prescription in 2016, and column (3) is the subset who did not have a gabapentin prescription in 2016. (A) Change in rate of patients per 1,000 with a Gabapentin prescription following classification in state to drug of concern for states that did not have mandatory access PDMP laws. This corresponds to the risk salience estimation shown in Equation 3. (B) Change in rate of patients per 1,000 with a Gabapentin prescription following classification in state to drug of concern, with a separate interaction to show the effect of having a mandatory access PDMP. This corresponds to the information effect estimation shown in Equation 4. (C) Change in rate of patients per 1,000 with a Gabapentin prescription following classification in state to Schedule V controlled substance. This corresponds to the hassle cost estimation shown in Equation 5.

*** p<0.01, ** p<0.05, * p<0.1

Appendix

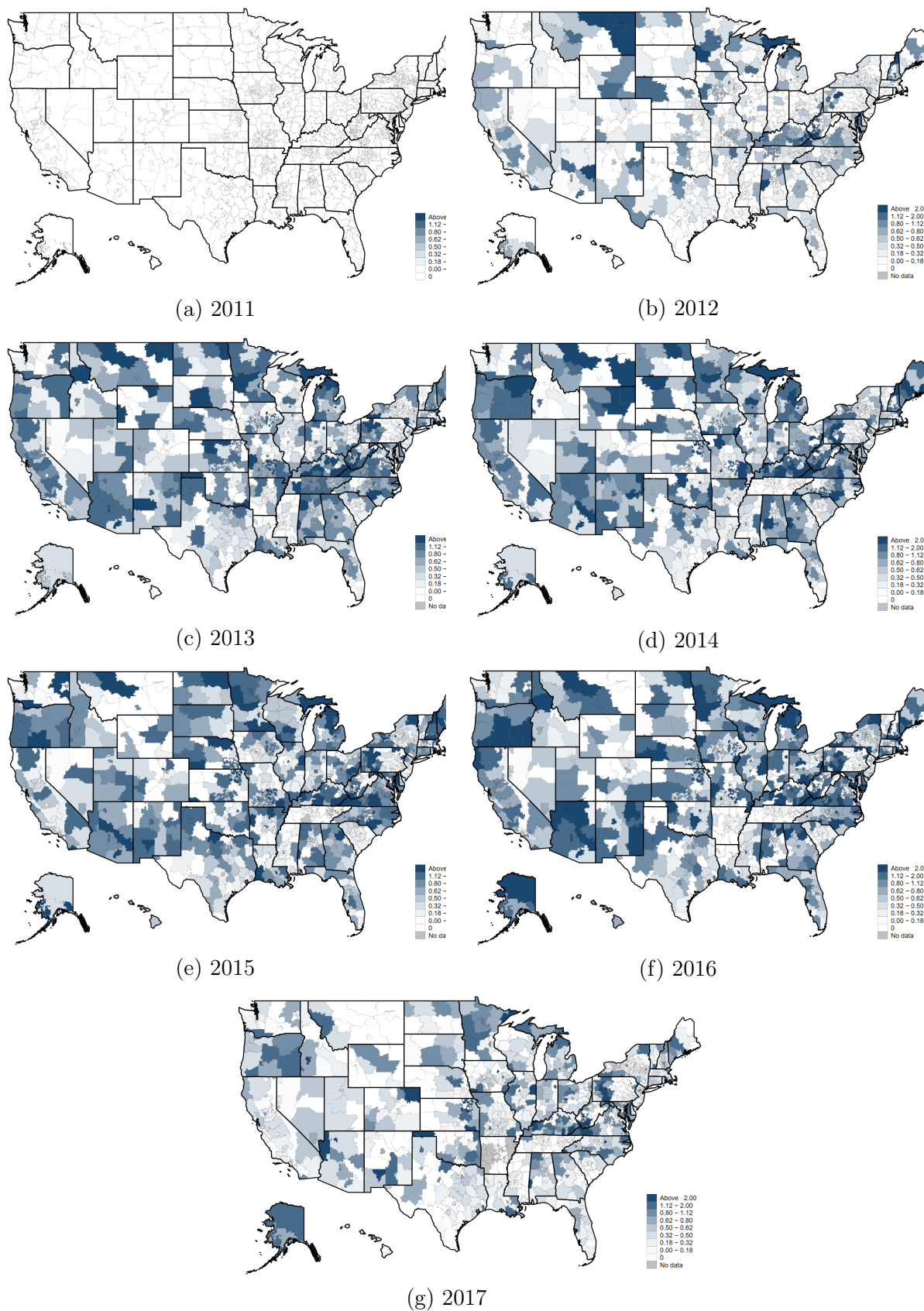
A Additional Descriptive Trends

Figure A1: Gabapentinoid Involved Deaths Over Time



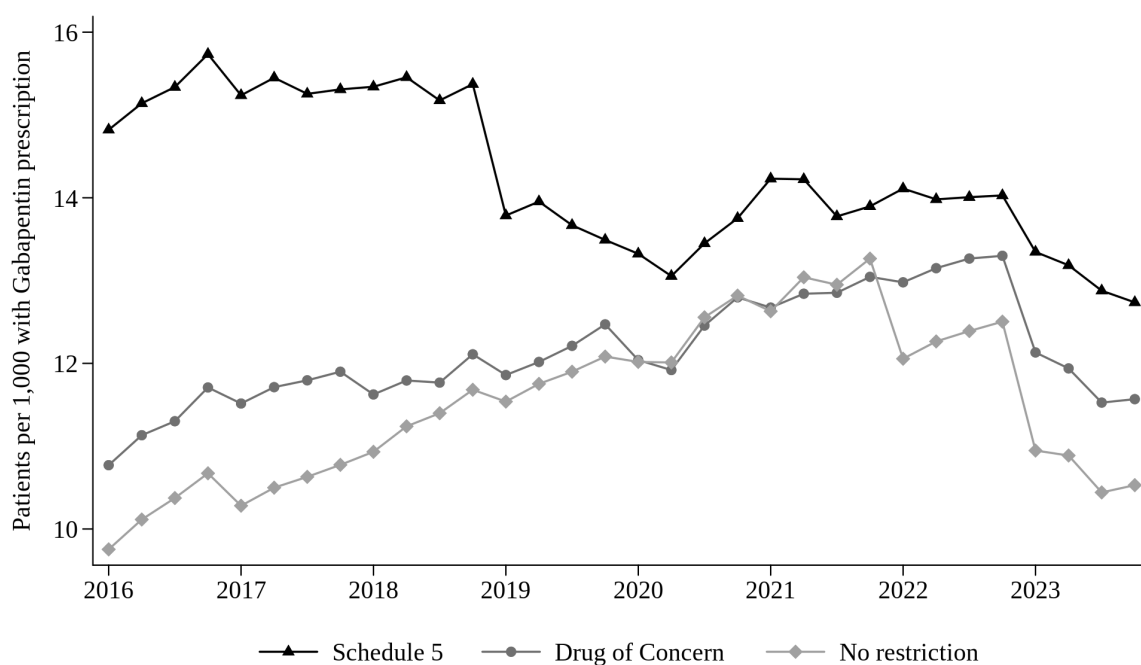
Notes: Annual average anticonvulsant and opioid involved mortality rates from the National Vital Statistics System (NVSS) from 2016-2023. In 2023, around 90% of Gabapentinoid involved deaths also involved an opioid, representing ~12% of opioid overdose involved deaths.

Figure A2: Poison Control Center Call Rates for Gabapentin-Involved Exposures per 100,000 Residents



Notes: Authors' calculations based on 3-digit ZIP code-level data from the National Poison Data System (2011–2017). Percentile cutoffs are anchored to 2013 call rates. 14.63% of ZIP codes recorded no gabapentin-related poison control calls in 2013. Among ZIP codes with at least one call, the legend reflects the 10th, 25th, 40th, 50th, 60th, 75th, and 90th percentiles of call volume.

Figure A3: Gabapentin Prescriptions by State Restriction Policy, No Base Year



Notes: Gabapentin prescriptions per 1,000 patients in MarketScan® commercial insurance claims data by state policy of ever classifying Gabapentin as a Schedule V controlled substance, ever as a Drug of Concern, or no restriction through the PDMP. Drug of concern policies add the drug to the state PDMP monitoring system without invoking the legal and administrative processes required for formal scheduling. Schedule-V policies trigger the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping.

Table A1: Gabapentin PDMP reporting policies

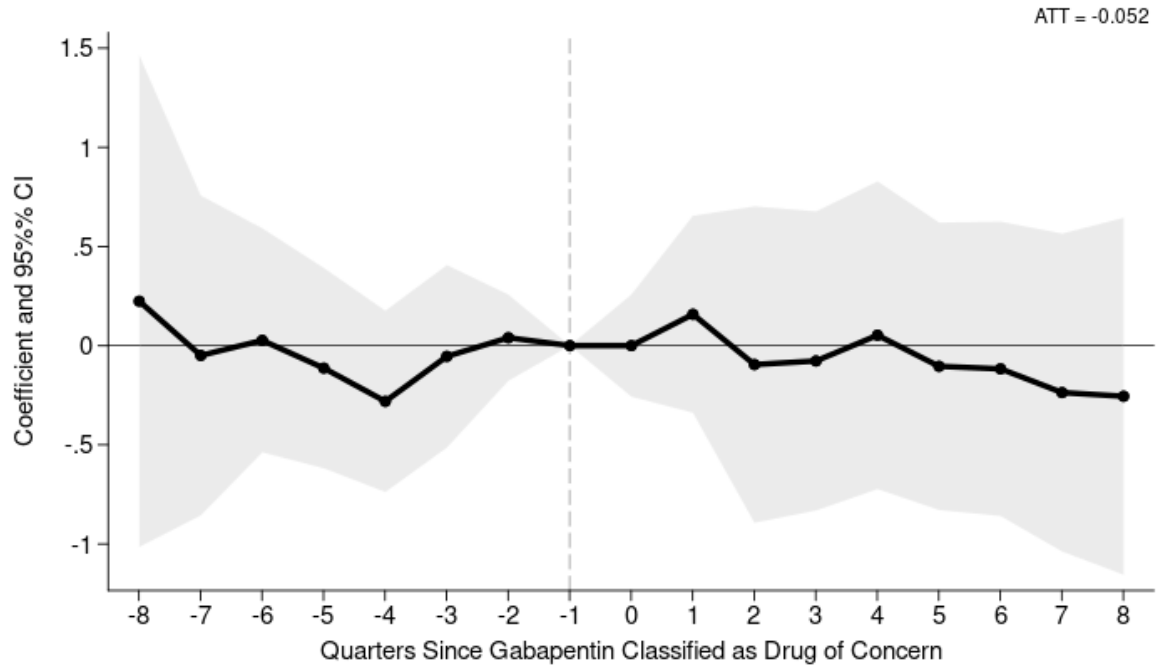
State/District	Date	Source
Scheduled as C-V and reported to PDMP		
Alabama	11/18/19	www.alabamapublichealth.gov/blog/assets/controlledsubstanceslist.pdf
Kentucky	3/3/17	https://pharmacy.ky.gov/Documents/Gabapentin%20-%20Schedule%20V%20Controlled%20Substance.pdf
Michigan	1/9/19	https://www.michigan.gov/opioids/nel/panel-news/news/gabapentin-scheduled-as-controlled-substance-to-help-with-states-opioid-epidemic
North Dakota	4/10/19	https://nabp.pharmacy/wp-content/uploads/2016/06/North-Dakota-Newsletter-June-2019.pdf
Tennessee	7/1/18	www.tn.gov/content/dam/tn/health/healthprofboards/New%20Statue%20Gabapentin%2006-18.pdf
Utah*	5/1/24	https://dopl.utah.gov/wp-content/uploads/2024/04/Gabapentin-Fact-Sheet-1.pdf
Virginia	7/1/19	https://legacylis.virginia.gov/cgi-bin/legp604.exe?191+sum+HB2557
West Virginia	9/5/18	www.wvlegislature.gov/bill_status/bills_history.cfm?input=4336&year=2018&sessiontype=RS&btype=bill
Drug of Concern reported to PDMP		
Connecticut	1/1/21	https://portal.ct.gov/-/media/DCP/drug_control/PMP/pdf/Gabapentin_Naloxone_CPMRS_2020.pdf
District of Columbia	6/7/19	https://dchealth.dc.gov/service/prescription-drug-monitoring-program
Illinois	7/7/23	www.ilga.gov/Commission/jcar/admincode/JCARTitlePart.asp?Title=077&Part=2081
Indiana	7/1/19	www.in.gov/pla/inspect/
Kansas	5/11/18	www.ktracs.ks.gov/home/showpublisheddocument/8408/638585193066700000
Louisiana	1/1/21	https://wwwcfprd.doa.louisiana.gov/boardsandcommissions/MeetingMinutes/336_2021-1013_Minutes_FinalPkg.pdf
Massachusetts	8/1/17	www.mass.gov/doc/requirements-for-gabapentin-data-submission-0/download
Minnesota	5/31/16	www.revisor.mn.gov/laws/2016/0/185/#laws.0.1.0
Mississippi	5/7/21	https://www.mbp.ms.gov/sites/default/files/inline-images/The%20Script%20VOL%209%20July%202021-2%20%282%29.pdf
New Jersey	5/6/19	www.njconsumeraffairs.gov/Adoptions/dirado_05062019.pdf
New Mexico	2/28/23	www.nmpmp.info/wp-content/uploads/2023/11/NM-PMP-Data-Submission-Dispenser-Guide_v4.0.pdf
North Dakota	8/1/17	https://nabp.pharmacy/wp-content/uploads/2016/06/North-Dakota-Newsletter-September-2017.pdf
Ohio	12/1/16	https://www.pharmacy.ohio.gov/documents/pubs/newsletter/2019/update%20on%20gabapentin%20in%20ohio.pdf
Oregon	1/1/20	https://www.oregon.gov/oha/PH/PreventionWellness/SafeLiving/PDMP/Documents/Oregon_PDMP_Data_Submission_Dispatcher_Guide.pdf
Utah	12/9/21	https://adminrules.utah.gov/public/rule/R156-37f/Current%20Rules
Virginia	2/23/17	https://legacylis.virginia.gov/cgi-bin/legp604.exe?171+ful+CHAP0181
Wisconsin	9/1/21	https://pdmp.wi.gov/statistics/controlled-substance-dispensing
Wyoming	7/1/17	https://wyoleg.gov/arules/2012/rules/ARR23-005.PDF

Notes: *Utah's schedule V classification occurred after our sample period and, thus, is considered only treated through its original drug of concern policy.

Nebraska is excluded from the study, as their state law required the PDMP to track all prescriptions.

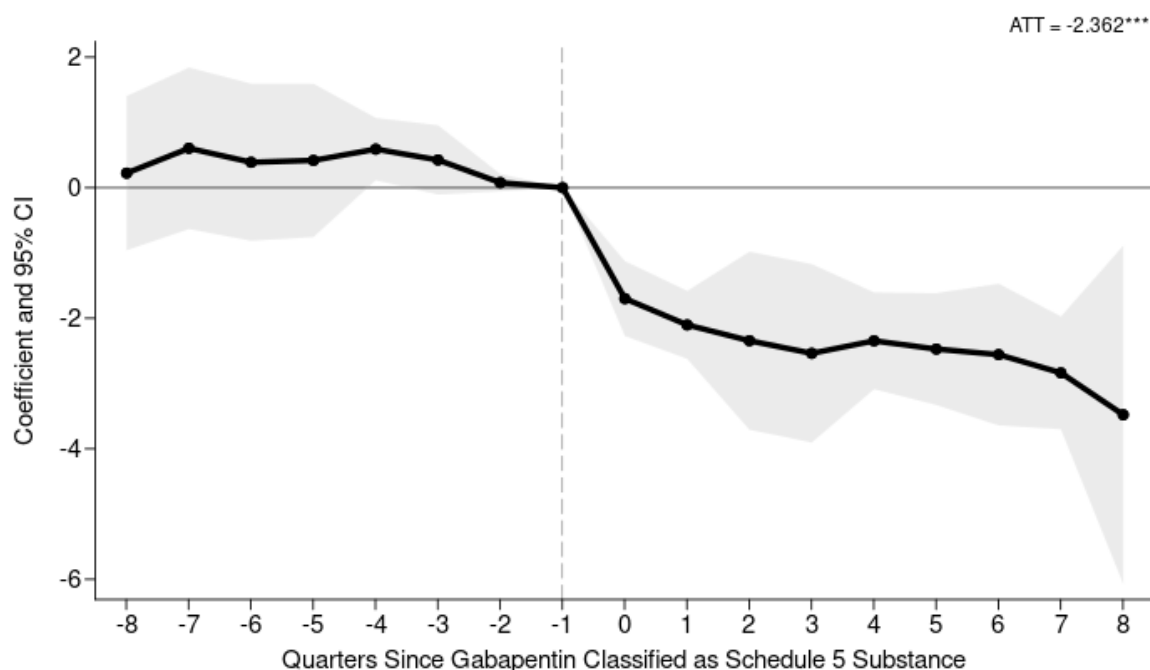
B Further Checks on the Effects of Gabapentin Regulation on Prescribing Behavior

Figure B1: Drugs of Concern Classification - Gabapentin Impact (Drop Schedule V)



Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Drug of Concern on Gabapentin prescription rates per 1,000 patients in MarketScan® commercial insurance claims data 2016-2023. Estimates for the 8 quarters before and after the policy change. Classification in state to Drug of Concern requires Gabapentin to be included in the state PDMP. This regression drops states who ever have a schedule V classification of Gabapentin, leaving only never treated states as the control group.

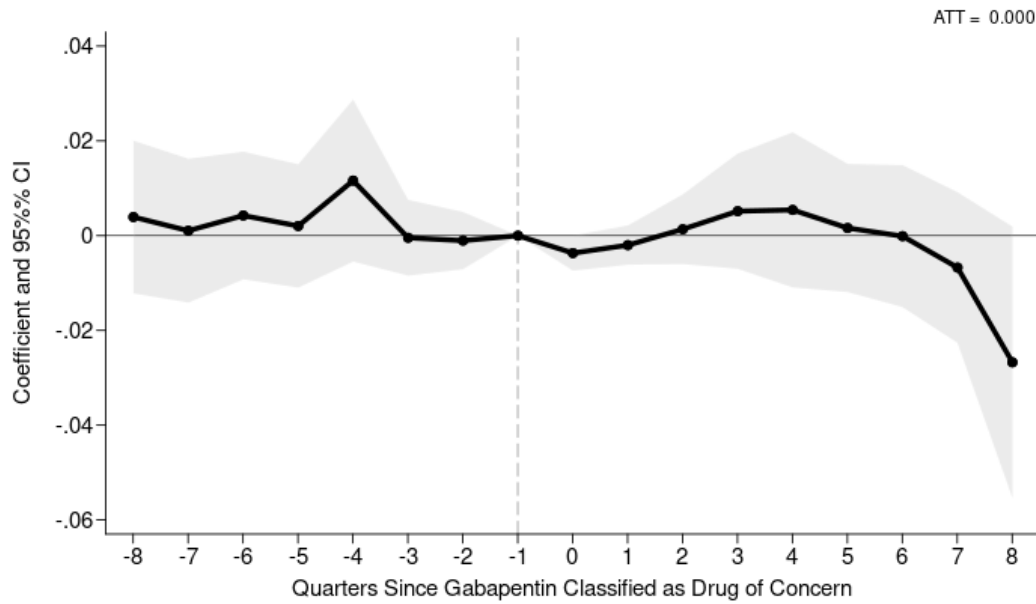
Figure B2: Schedule V Classification - Gabapentin Impact (Drop Drugs of Concern)



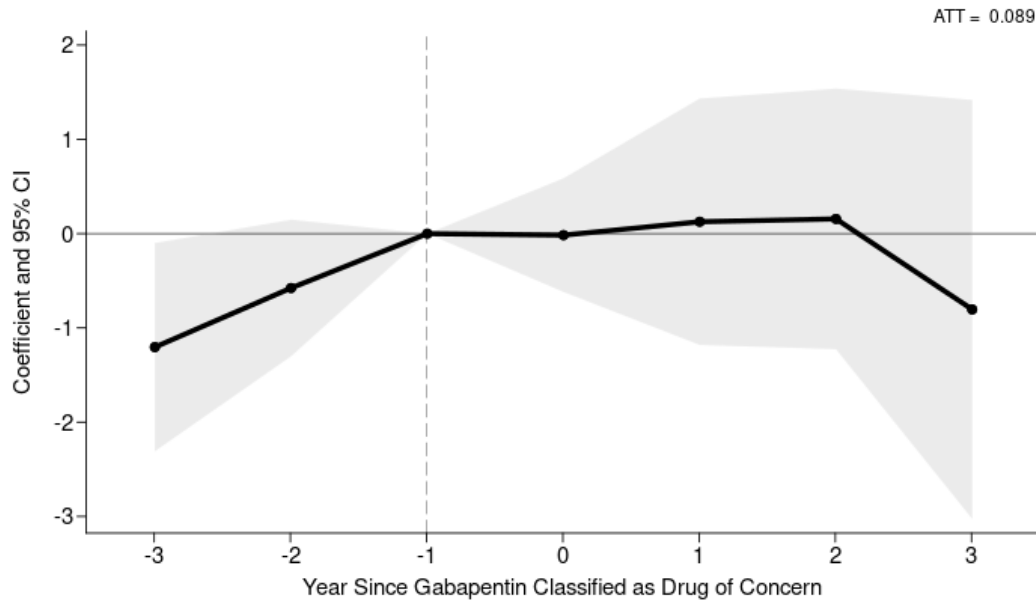
Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on Gabapentin prescription rates per 1,000 patients in MarketScan® commercial insurance claims data 2016-2023. Being a schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Estimates for the 8 quarters before and after the policy change. This regression drops states who only ever have a Drug of Concern classification of Gabapentin, leaving only never treated states as the control group.

Figure B3: Gabapentin: Drug of Concern Classification - Public Payer

(a) Medicaid



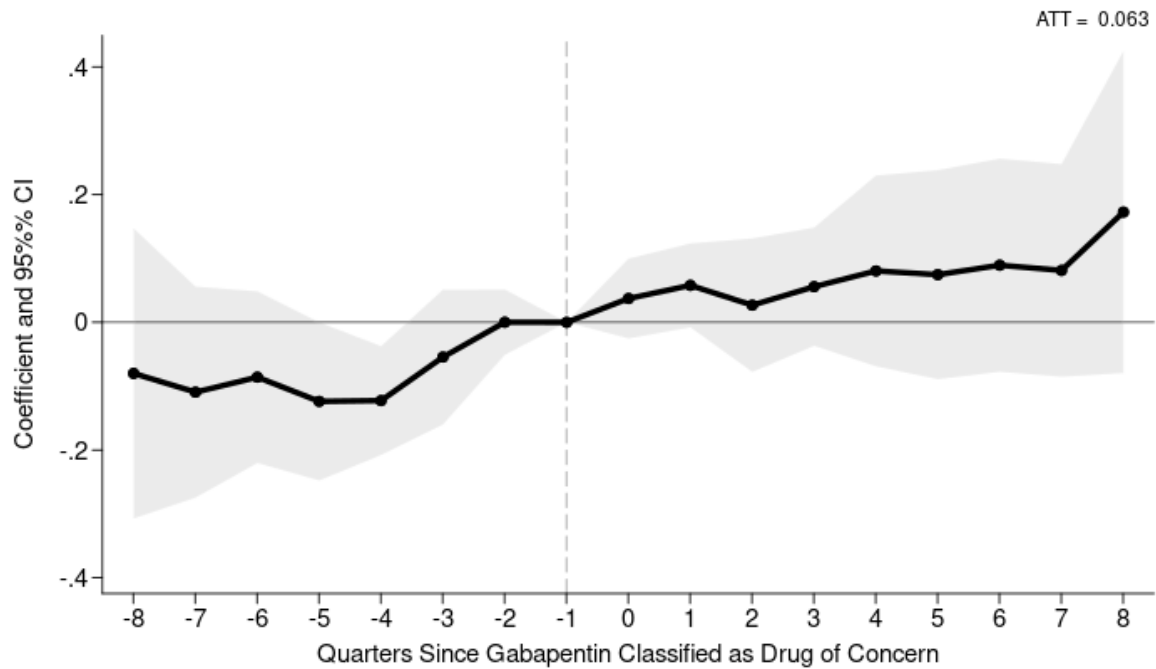
(b) Medicare



Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate impacts of classifying Gabapentin as a Drug of Concern on Gabapentin prescription rates for patients using publicly funded insurances: Medicaid and Medicare. c (a) Change in rate of rate of Medicaid enrollees per 1,000 with a Gabapentin prescription following Drug of Concern classification in state. Estimates for the 8 quarters before and after the policy change. Control group is never treated and Schedule V states. (b) Change in rate of rate of Medicare Part D enrollees per 1,000 for an average quarter each year with a Gabapentin prescription following Drug of Concern classification in state. Estimates for the 3 years before and after the policy change.

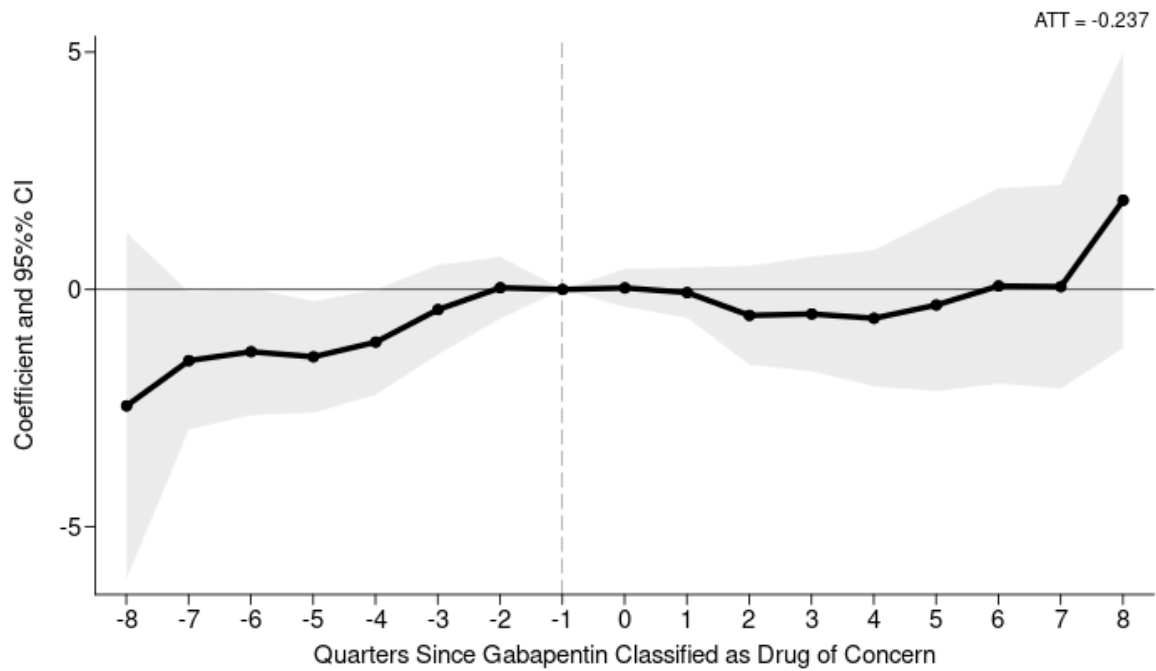
C Substitution Toward Other Pharmaceutical and Non-Pharmaceutical Pain Management

Figure C1: Drugs of Concern Classification - Pregabalin Impact



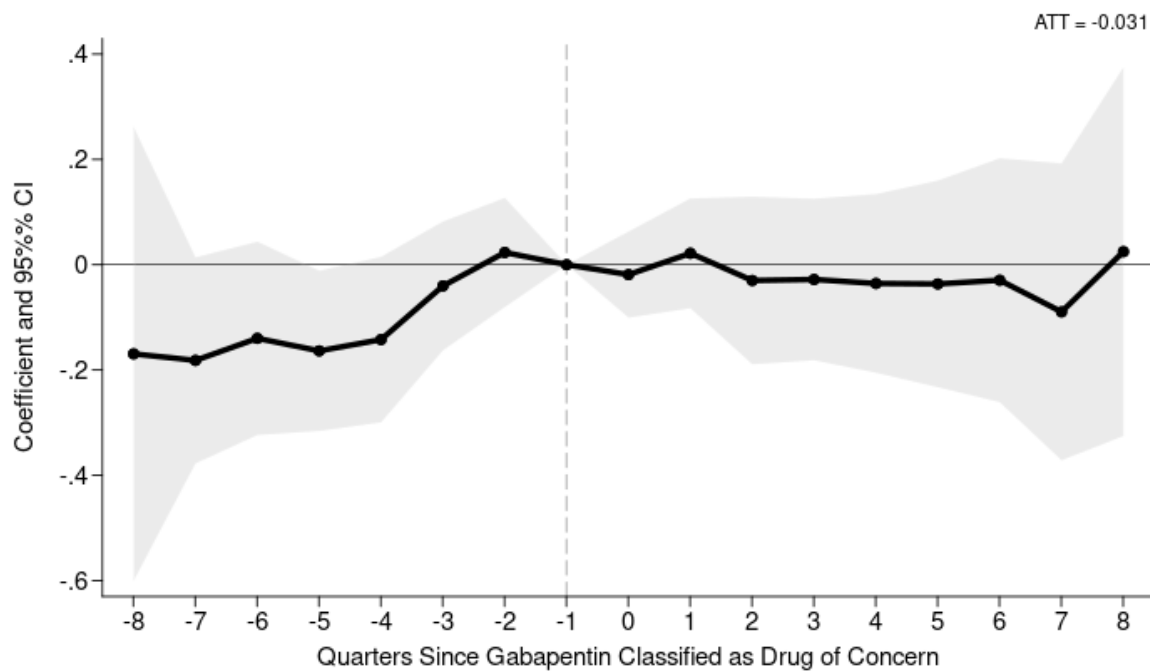
Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Drug of Concern on prescription rates of Pregabalin – a potential substitute for Gabapentin – per 1,000 patients in MarketScan® commercial insurance claims data 2016-2023. Classification in state to Drug of Concern requires Gabapentin to be included in the state PDMP. Estimates for the 8 quarters before and after the policy change. Control group is never treated and Schedule V states.

Figure C2: Drugs of Concern Classification - Opioid Impact



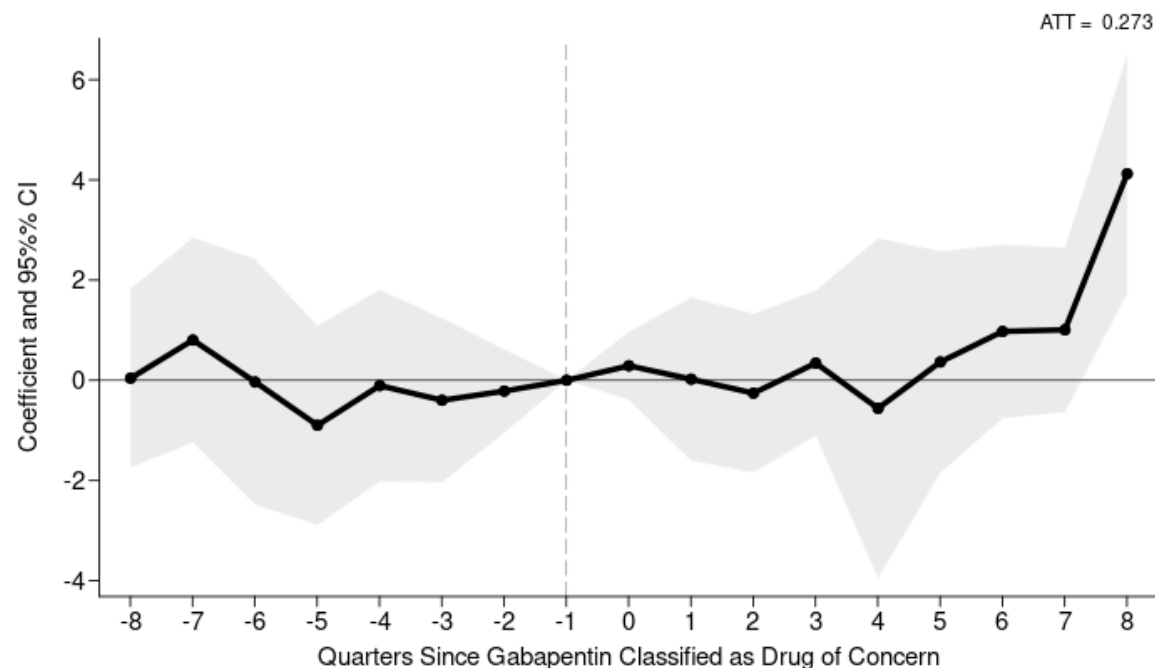
Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Drug of Concern on prescription rates of opioids per 1,000 patients in MarketScan® commercial insurance claims data 2016-2023. Classification in state to Drug of Concern requires Gabapentin to be included in the state PDMP. Estimates for the 8 quarters before and after the policy change. Control group is never treated and Schedule V states.

Figure C3: Drugs of Concern Classification - Co prescriptions of Gabapentin and Opioids



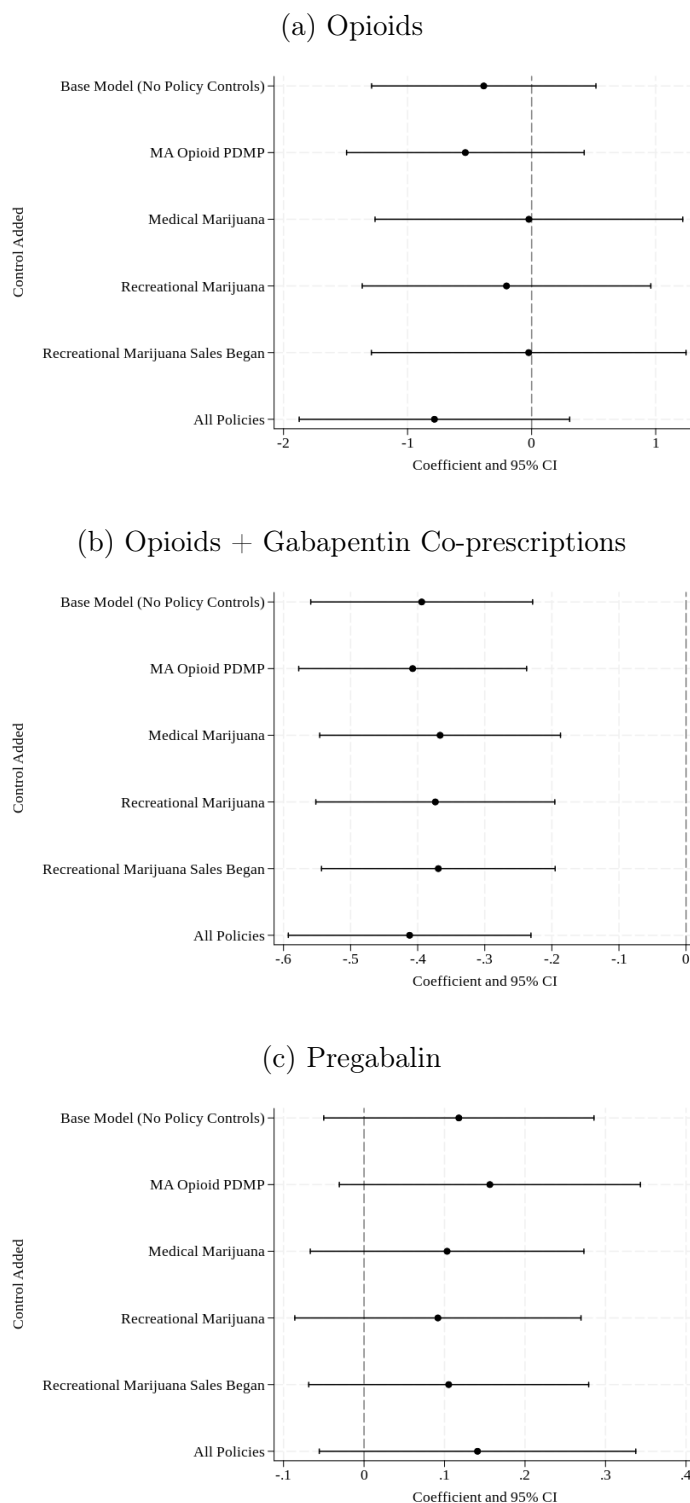
Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Drug of Concern on co-prescription rates of Gabapentin and an opioid per 1,000 patients in MarketScan® commercial insurance claims data 2016-2023. Classification in state to Drug of Concern requires Gabapentin to be included in the state PDMP. Estimates for the 8 quarters before and after the policy change. Control group is never treated and Schedule V states.

Figure C4: Drugs of Concern Classification - Non-Pharmacological Pain Management



Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Drug of Concern on rates of patients per 1,000 with a non-pharmacological pain management procedure among MarketScan® commercial insurance claims data 2016-2023. Classification in state to Drug of Concern requires Gabapentin to be included in the state PDMP. Estimates for the 8 quarters before and after the policy change. Control group is never treated and Schedule V states.

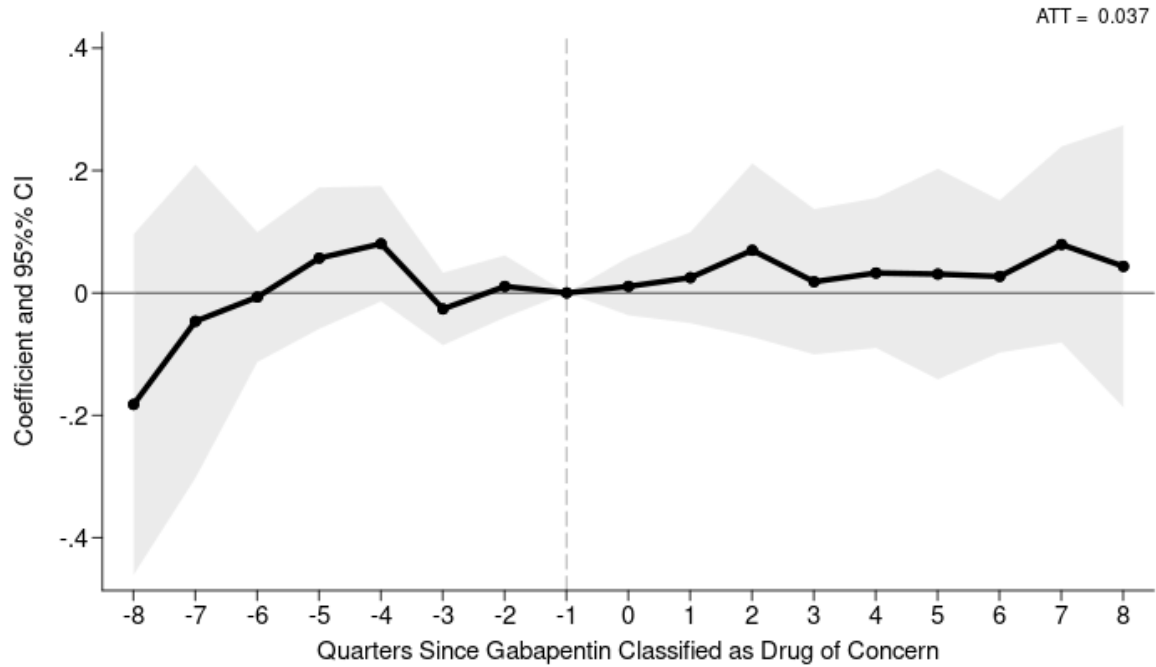
Figure C5: Potential Substitutes: Schedule V Classification Classification - Other policies
Robustness Check



Notes: Average treatment on the treated effect (ATT) results using a stacked difference-in-differences model from Equation 2 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on prescription rates per 1,000 patients in MarketScan® commercial insurance claims data 2016-2023. Estimates based on the 8 quarters before and after the policy change. Being a Schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Each estimate below the base model includes control variables for other policies which may impact prescribing of gabapentin via its substitutes. MA Opioid PDMP policies require prescribers of opioids to consult the state PDMP prior to issues prescriptions. Medical Marijuana policies legalize the medical use and sale of marijuana and products which contain it. Recreational Marijuana policies legalize the use and sale of marijuana and related products without the need for medical reasoning nor a prescription. Recreational sales in certified establishments often began after formal recreational legalization, making these later dates the timing when individuals may have purchased the products from a shop. The bottom estimate includes controls for all policies. Panel (a) shows the impact on opioid prescription rates, Panel (b) shows the impact on opioid and gabapentin co-prescription rates, and Panel (c) shows the impact on pregabalin prescription rates, all following classification in state to Schedule V controlled substance.

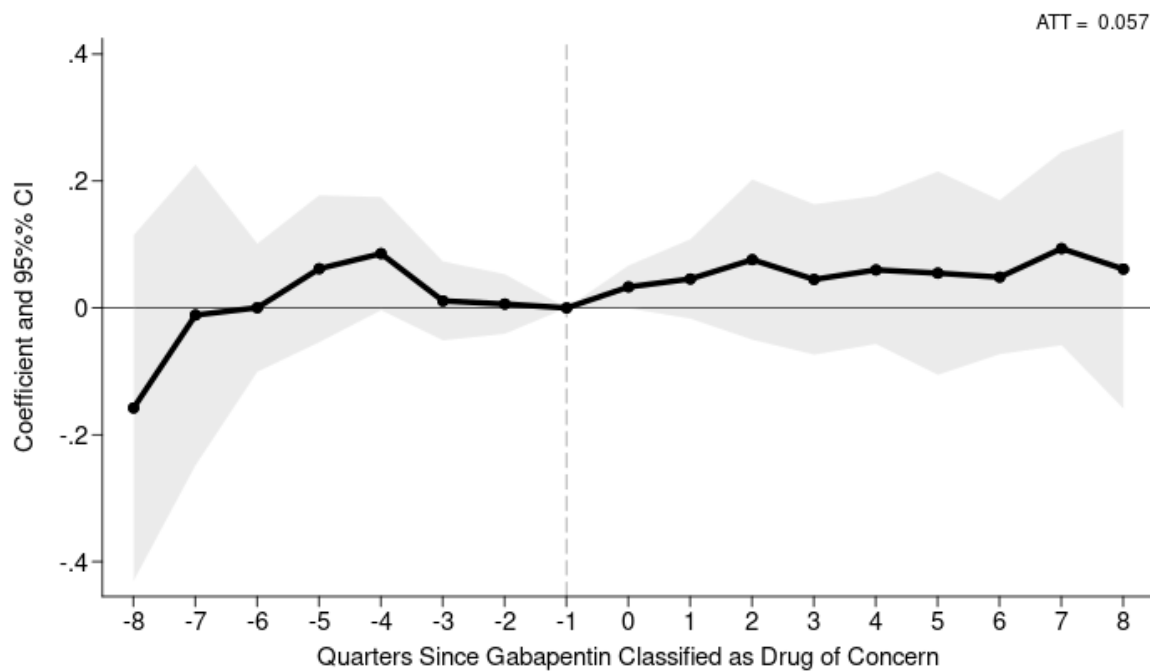
D Additional Evidence on Fatal and Non-Fatal Gabapentin-Involved Adverse Events

Figure D1: Drugs of Concern Classification - Fatal Events Involving Gabapentin



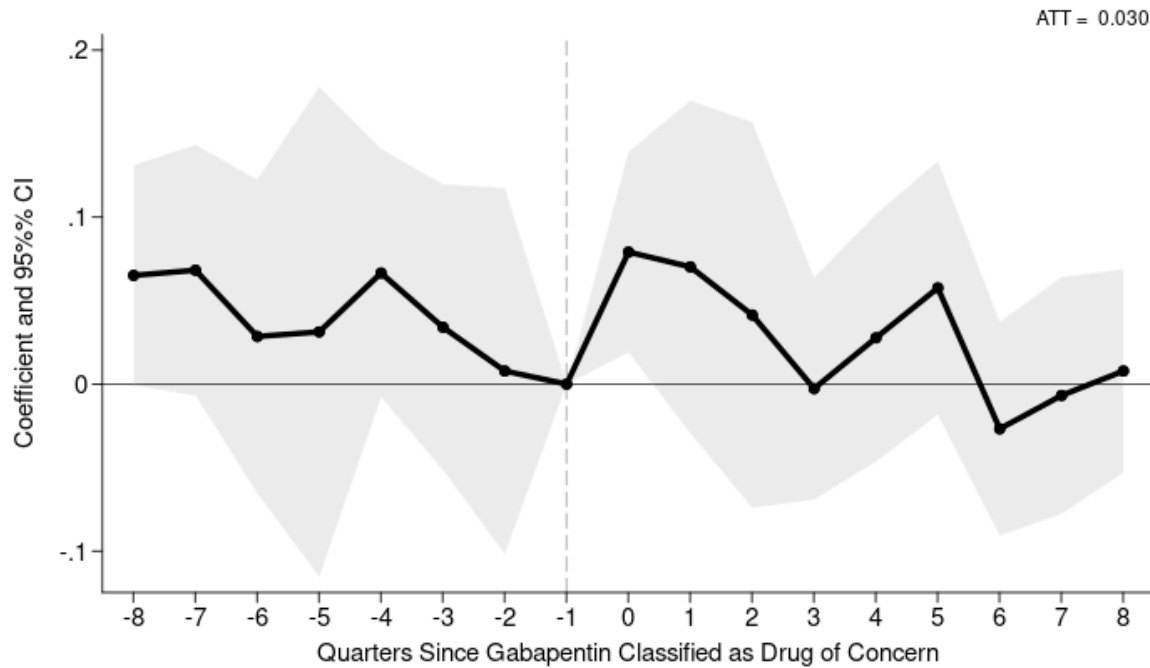
Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Drug of Concern on anticonvulsant involved mortality rates from the National Vital Statistics System (NVSS) from 2016-2023. Classification in state to Drug of Concern requires Gabapentin to be included in the state PDMP. Estimates for the 8 quarters before and after the policy change. Control group is never treated and Schedule V states.

Figure D2: Drugs of Concern Classification - Fatal Events Involving Gabapentin and Opioids



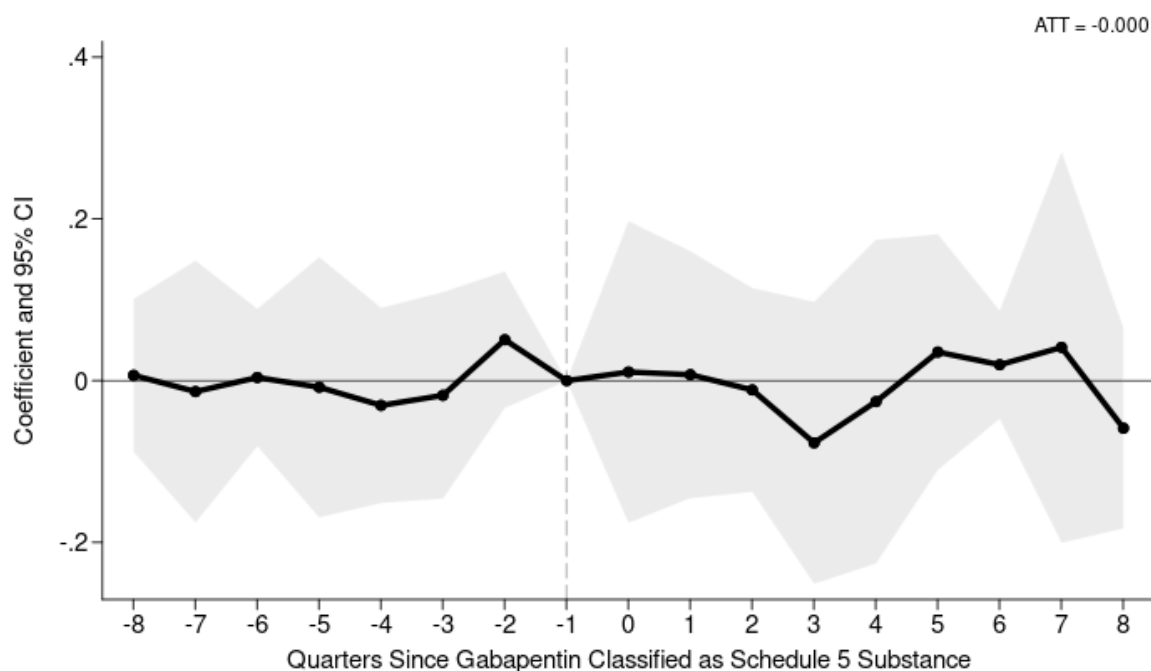
Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Drug of Concern on anticonvulsant and opioid involved mortality rates from the National Vital Statistics System (NVSS) from 2016-2023. Classification in state to Drug of Concern requires Gabapentin to be included in the state PDMP. Estimates for the 8 quarters before and after the policy change. Control group is never treated and Schedule V states.

Figure D3: Drugs of Concern Classification - Non-fatal Events Involving Gabapentin



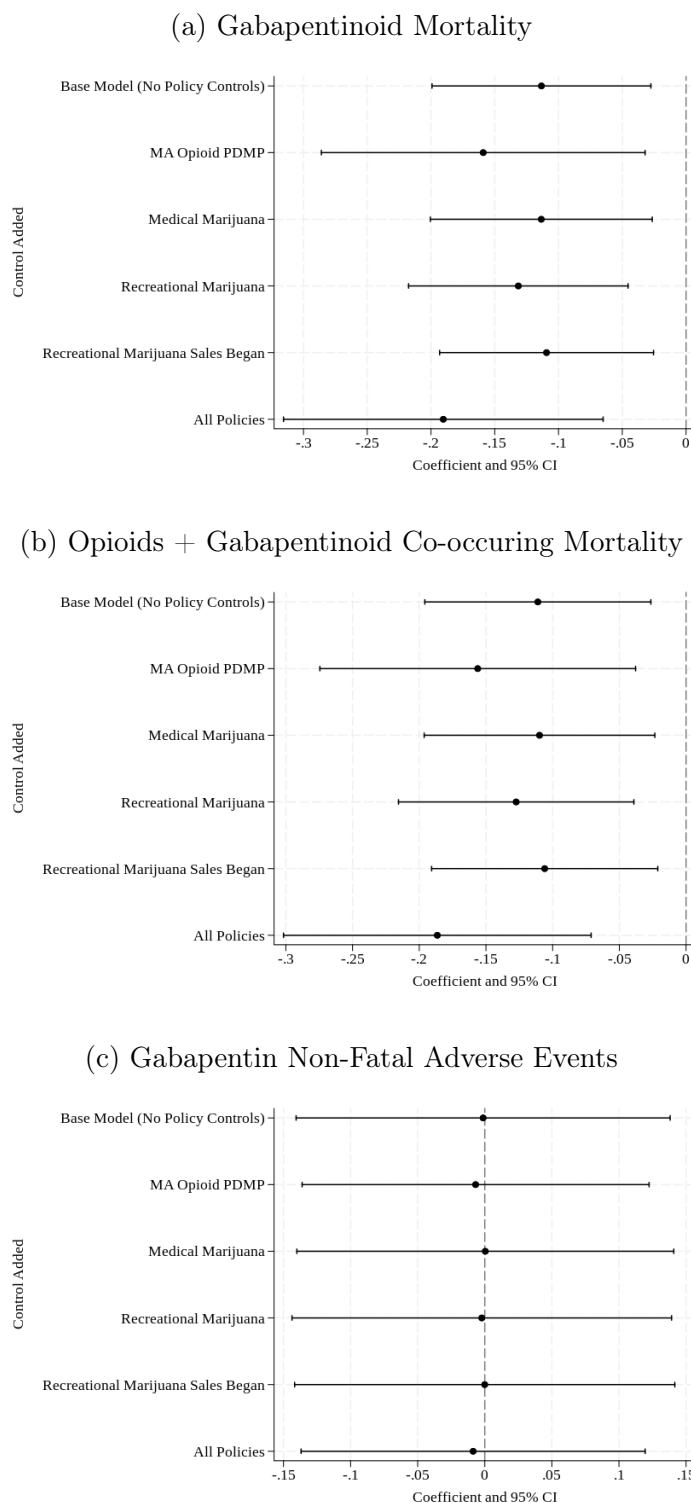
Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Drug of Concern on incidence rate per 100,000 patients of non-fatal events including anticonvulsants among MarketScan® commercial insurance claims data 2016-2023. Classification in state to Drug of Concern requires Gabapentin to be included in the state PDMP. Estimates for the 8 quarters before and after the policy change. Control group is never treated and Schedule V states.

Figure D4: Schedule V Classification - Non-fatal Events Involving Gabapentin



Notes: Event study results using a stacked difference-in-differences model from Equation 1 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on incidence rate per 100,000 patients of non-fatal events including anticonvulsants among MarketScan® commercial insurance claims data 2016-2023. Being a Schedule V controlled substance triggers the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced record keeping. Estimates for the 8 quarters before and after the policy change. Control group is never treated and DOC states.

Figure D5: Fatal and Non-fatal Adverse Events: Schedule V Classification Classification
- Other policies Robustness Check



Notes: Average treatment on the treated effect (ATT) results using a stacked difference-in-differences model from Equation 2 to estimate the impacts of classifying Gabapentin as a Schedule-V controlled substance on anticonvulsant involved adverse event rates. Estimates based on the 8 quarters before and after the policy change. Classification in state to Drug of Concern requires Gabapentin to be included in the state PDMP. Each estimate below the base model includes control variables for other policies which may impact prescribing of gabapentin via its substitutes. MA Opioid PDMP policies require prescribers of opioids to consult the state PDMP prior to issues prescriptions. Medical Marijuana policies legalize the medical use and sale of marijuana and products which contain it. Recreational Marijuana policies legalize the use and sale of marijuana and related products without the need for medical reasoning nor a prescription. Recreational sales in certified establishments often began after formal recreational legalization, making these later dates the timing when individuals may have purchased the products from a shop. The bottom estimate includes controls for all policies. Panel (a) shows the impact on gabapentinoid involved mortality rates per 100,000 population, Panel (b) shows the impact on gabapentinoid and opioid involved mortality rates per 100,000 population, and Panel (c) shows the impact on incidence rate per 100,000 patients of non-fatal events including anticonvulsants among MarketScan® commercial insurance claims data.